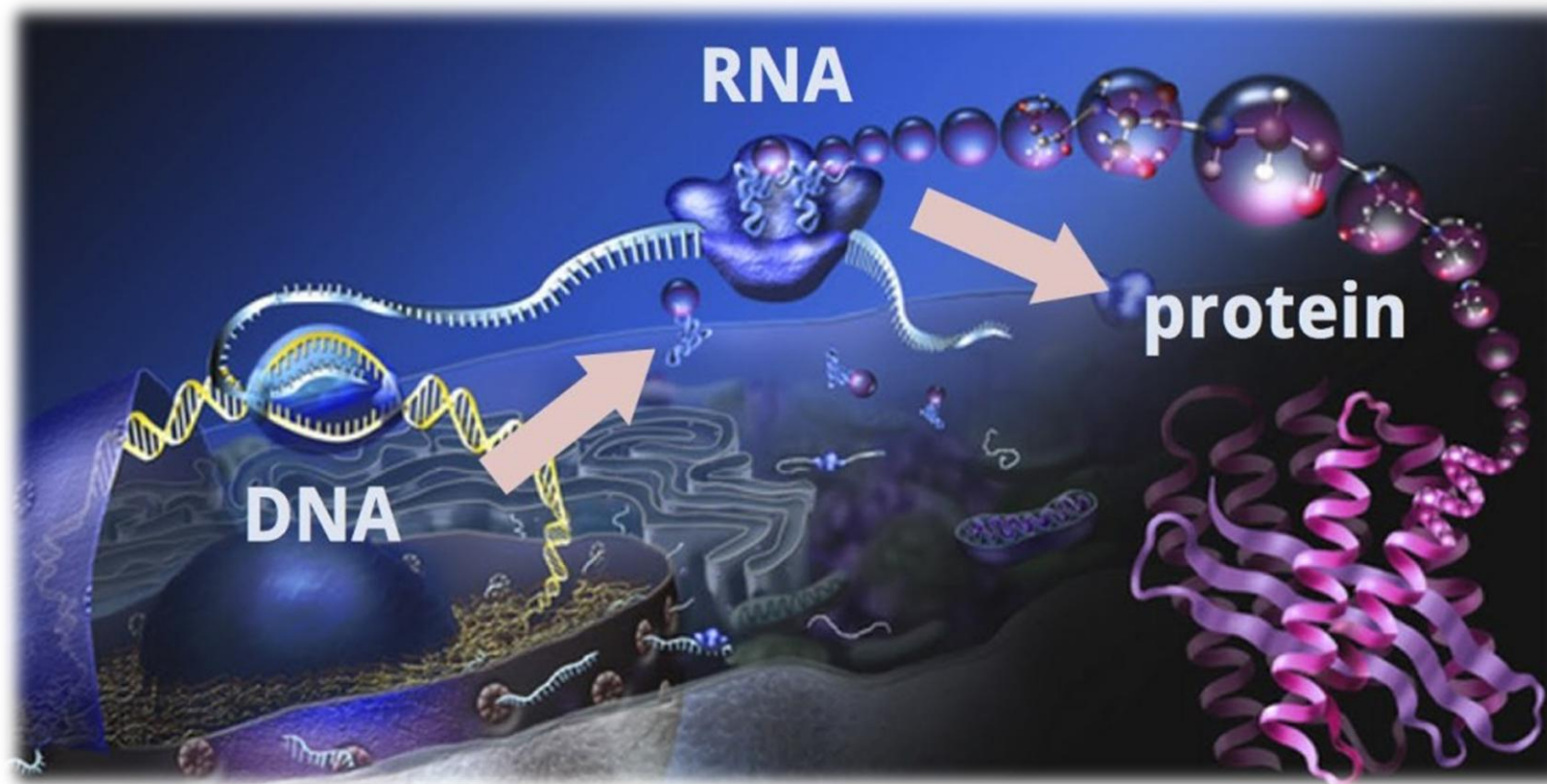


ANALYSIS OF TRANSCRIPTOMES



Rubén Sancho; Bruno Contreras-Moreira

16 December 2025 (6 hours)

9:00-10:00

10:00-11:00

11:30-13:30

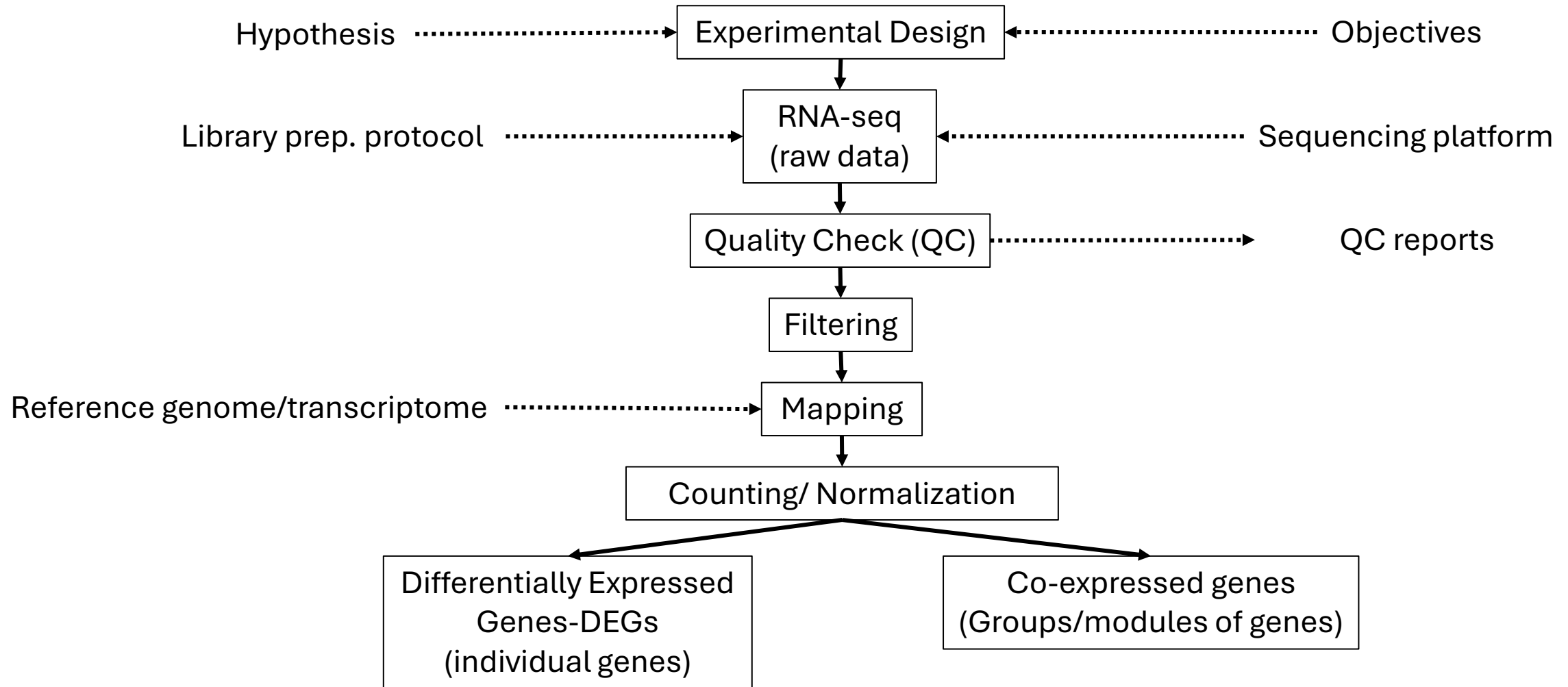
15:00-17:00

Session Objectives

- ✓ Gain an overview of the complete RNA-seq analysis workflow.
- ✓ Understand differential gene expression (DEG) and co-expression network analyses.
- ✓ Understand the key concepts underlying these analyses.
- ✓ Become familiar with the bioinformatics tools and processes used in RNA-seq analysis.

How to analyze RNA-seq data?

From raw data to interpreting differential expression results, co-expression networks, and biological implications.



Experimental Design & RNA-seq

Experimental design: ✓ Tissues ✓ Samples ✓ Reps ✓ Treatments ✓ Sampling time

Library type: ✓ mRNA-seq (PolyA+) ✓ Total RNA-seq (no rRNA) ✓ Strand-specific RNA-seq ✓ Small RNA-seq ✓ scRNA-seq ✓ 3'-end RNA-seq

*The most widely used protocol involves the **TruSeq** (Illumina) kit, which assumes abundant starting material (0.1–1 µg total RNA) and maintains strand specificity*

Sequencing (read) type: ✓ Single-End ✓ Paired-end ✓ Short ✓ Long

Goals: ✓ Differentially Expressed Genes ✓ Co-expressed genes ✓ Isoform/Gene level ...

The choice of bioinformatics tools and commands depends on this information!!!

RNA-seq Library Types

Type	Preparation	Captured RNA	Advantages	Disadvantages	Applications
mRNA-seq (PolyA+)	Selection of polyadenylated RNA using oligo(dT) beads	Mature mRNA (polyA+)	High specificity for coding genes ; reproducible; high proportion of informative reads	Does not detect noncoding or non-polyA RNAs; unsuitable for degraded RNA	Differential gene expression; standard transcriptome profiling
Total RNA-seq	Ribosomal RNA removal (Ribo-Zero, NEBNext, etc.)	All RNAs except rRNA	Captures coding and noncoding RNAs ; works with degraded RNA	Higher background noise; more expensive	Whole transcriptome profiling; lncRNA studies; FFPE* samples
Strand-specific RNA-seq	Incorporation of strand-specific markers during cDNA synthesis	Same as mRNA-seq or Total RNA-seq, but retains strand orientation	Distinguishes sense/antisense transcripts; improves isoform annotation	Slightly higher cost and complexity	Antisense RNA, isoform detection, transcriptional regulation
Small RNA-seq	Size selection (18–30 nt fragments)	miRNA, siRNA, piRNA, small snRNA**	Enables post-transcriptional regulation studies	Does not detect mRNA or long transcripts	miRNA profiling, piRNA discovery, small RNA pathways
3'-end RNA-seq (TagSeq, QuantSeq)	Captures only the 3' ends of transcripts using short oligo(dT) priming	3' end of mRNA	Cost-effective; simple; correlates well with gene expression	No isoform or splicing information	Rapid expression quantification; high-throughput screening
Single-cell RNA-seq (scRNA-seq)	Cell barcoding using droplets or nanowells	Total RNA from individual cells	Single-cell resolution; reveals cellular heterogeneity	Expensive; lower coverage per cell; complex analysis	Developmental biology; cancer heterogeneity; immune profiling
Exome-capture RNA-seq	Hybridization with probes targeting known exons	Exonic coding regions	Focused on genes of interest ; reduces background	Misses noncoding RNAs and unannotated isoforms	Targeted studies; validation of candidate genes

Quality Check (QC)

In the Lab: concentrations, absorbances and integrity (RIN- RNA Integrity Number 1-10)

	Optimal Value	Minimum	Comments
A260/A280	1.8–2.1	≥ 1.7	Low values indicate protein or phenol contamination.
A260/A230	>2.0	≥ 1.8	Low values suggest salt or reagent contamination.
Concentration	100 ng – 2 μ g total RNA	≥ 50 ng	Depends on protocol; higher input improves library complexity.
RIN (RNA Integrity Number)	≥ 8	≥ 6 (for rRNA-depleted RNA-seq)	Below 6, use specialized protocols (e.g., 3'-end RNA-seq).

In silico (check QC reports): Per base sequence quality; Per tile sequence quality; Per sequence quality scores; Per base sequence content; Per sequence GC content; Per base N content; Sequence Length Distribution; Sequence Duplication Levels; Overrepresented sequences; Adapter Content

Filtering

Remove adapters and low-quality sequences and/or nucleotides

Trimmomatic software

- ILLUMINACLIP: Cut adapter and other Illumina-specific sequences from the read.
- SLIDINGWINDOW: Perform a sliding window trimming, cutting once the average quality within the window falls below a threshold.
- LEADING: Cut bases off the start of a read, if below a threshold quality
- TRAILING: Cut bases off the end of a read, if below a threshold quality
- CROP: Cut the read to a specified length
- HEADCROP: Cut the specified number of bases from the start of the read
- MINLEN: Drop the read if it is below a specified length
- TOPHRED33: Convert quality scores to Phred-33
- TOPHRED64: Convert quality scores to Phred-64

Mapping and Counting

Mapping or aligning the sequences against a reference genome/transcriptome is a key step in identifying which genes the sequenced RNA fragments originate from and thus quantifying their expression.

To select the alignment/mapping software we must take into account:

- ✓ What reference is available, genome or transcriptome?
- ✓ What downstream analyses will be performed: differential expression, co-expression networks, ...?

Software	Alignment type	Reference	Speed	Accuracy	Splicing detection	Direct quantification	Output	Key notes
STAR 	<i>Full alignment (sequence-based)</i>	Genome	Very high	Very high	Yes	No (use featureCounts or RSEM)	SAM/BAM files	Gold standard in large projects (e.g., ENCODE, GTEx); excellent for novel genes or isoforms
HISAT2 	<i>Full alignment (hierarchical indexing)</i>	Genome	High	High	Yes	No (use StringTie or featureCounts)	SAM/BAM files	Fast and memory-efficient; successor to TopHat2
Salmon 	<i>Quasi-mapping</i>	Transcriptome	Very high	High	Not directly (depends on annotation)	Yes	Expression tables (TPM, FPKM, counts)	Very fast; ideal for direct quantification without BAM files
Kallisto 	<i>Pseudoalignment</i>	Transcriptome	Extremely high	High	No	Yes	Expression tables (TPM, counts)	Ultra-fast; great for exploratory or large-cohort analyses

Counting, Normalization and Differential Expression analysis



Gene Expression Measurement

Raw counts: Number of sequencing reads mapped to each gene.

Cannot be directly compared between genes or samples:

- Longer genes = more sequenced reads
- Samples with deeper sequencing = more total sequenced reads

**Normalize for
sequencing
depth and
gene length**

RPKM: Reads Per Kilobase of exon per Million reads mapped. **Single-end** (SE) RNA-seq.

FPKM: Analogous to RPKM and is used specifically in **paired-end** (PE) RNA-seq experiments.

$$RPKM_i \text{ or } FPKM_i = \frac{q_i}{\frac{l_i}{10^3} * \frac{\sum_j q_j}{10^6}} = \frac{q_i}{l_i * \sum_j q_j} * 10^9$$

$q_i \rightarrow$ raw read or fragment counts (per feature and sample)

$l_i \rightarrow$ feature (i.e., gene or transcript) length (per feature, bp)

$\sum_j q_j \rightarrow$ total number of mapped reads or fragments (per sample)

TPM: Transcript Per Million.

The sum of all TPM values is the same in all samples. A TPM value represents a relative expression level that should be comparable between samples.

$$TPM_i = \frac{q_i / l_i}{\sum_j (q_j / l_j)} * 10^6$$

$q_i \rightarrow$ reads mapped to transcript (per transcript and sample)

$l_i \rightarrow$ transcript length (per transcript bp)

$\sum_j (q_j / l_j) \rightarrow$ the sum of mapped reads to transcript normalized by transcript length

Tools for Counting/Normalization and Differential Expression analysis

Tool	Type	Input	Output	Level	Use	Advantages	Disadvantages
featureCounts 	Counting	Aligned reads (BAM/SAM)	Raw counts	Gene	Read summarization	Fast, memory-efficient	No isoform quantification
RSEM 	Quantification	Aligned reads (or raw FASTQ)	TPM, FPKM, counts	Gene & transcript	Quantifying expression	Handles multi-mapping reads, isoform-aware	Slow, computationally demanding
StringTie 	Assembly + Quantification	Aligned reads	Assembled transcripts + FPKM	Transcript	Transcript reconstruction	Detect novel isoforms	Less accurate with low coverage
DESeq2 	Differential expression	Raw counts	Fold change, p-values	Gene	Expression comparison	Statistically robust, well-documented	Requires raw counts, not TPM
Sleuth 	Differential expression	TPMs (from Kallisto)	Fold change, q-values	Transcript	Fast DE analysis	Very fast, interactive, works with Kallisto	Kallisto input

Differentially Expressed Genes (DEGs) using Sleuth

Genes whose expression levels significantly increase or decrease between different conditions or groups

Different tissues (leaves, seeds, flower parts, roots, ...), different growth stages, treatments/doses (e.g. nitrogen, ...), abiotic stresses (heat, cold, drought, waterlogging, nutrient deficiency, ...), or biotic stresses (diseases, ...).



Sleuth is used to explore gene expression results from Kallisto and to perform differential expression analysis.

Sleuth is implemented in an interactive shiny app that utilizes Kallisto quantifications and bootstraps for fast and accurate analysis of data from RNA-seq experiments.

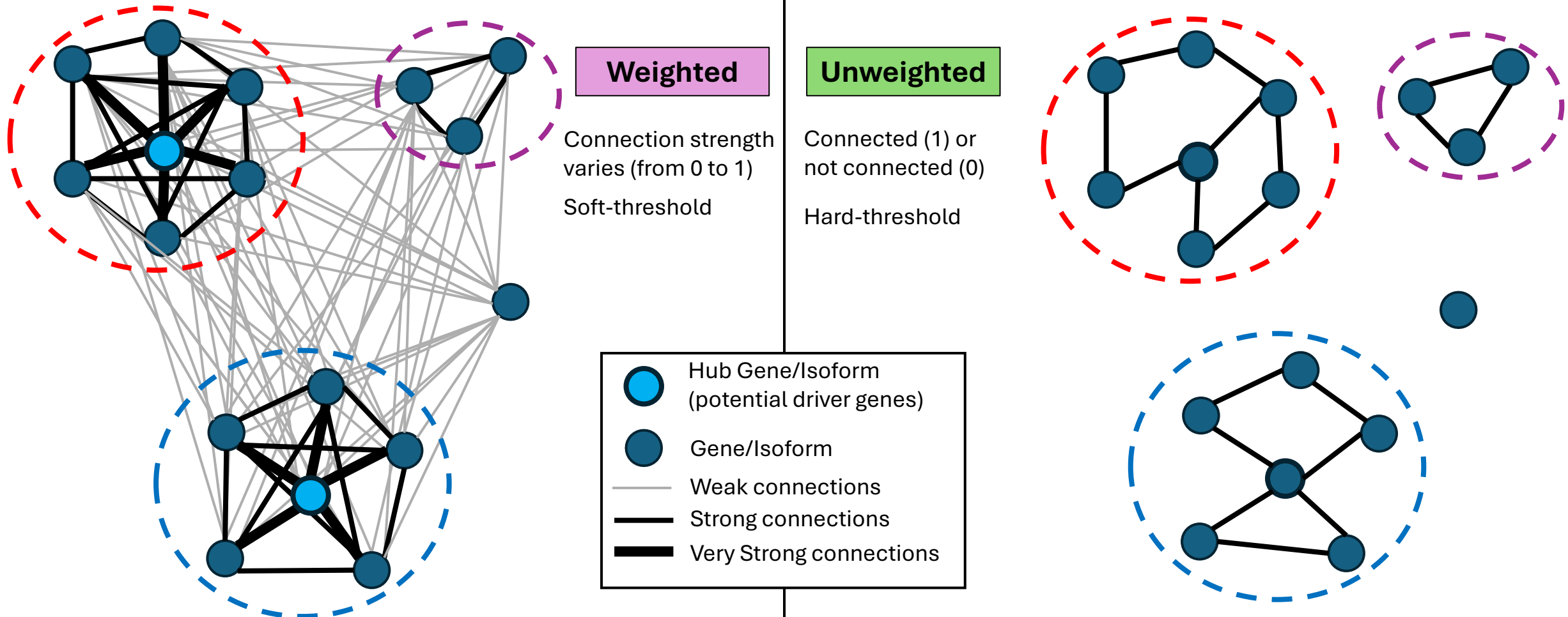
Sleuth uses the bootstraps to ascertain and correct for technical variation in experiments.

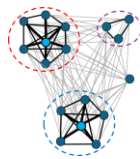
Gene Co-expression Networks

Graphical representations that show how genes are related based on the **similarity of their expression patterns**.

● **Node = Gene** — **Edge = Connection (correlated expression levels)**

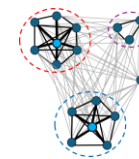
***Premise:** Genes with similar expression patterns (co-expressed) are functionally associated*





WGCNA

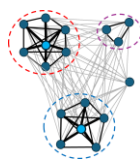
Weighted Gene Co-expression Network Analysis



Glossary of WGCNA Terminology

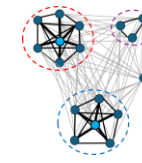
1. Data set

- ✓ Expression matrix (normalized expression values) with rows as genes/isoforms and columns as samples.
- ✓ Not filtering genes by differential expression.
- ✓ It is recommended that the analysis be performed with a larger number of samples, at least 20.
- ✓ Clean the data by removing obvious outlier samples as well as genes and samples with excessive numbers of missing entries.



WGCNA

Weighted Gene Co-expression Network Analysis



2. Weighted Network construction

Adjacency: matrix $n \times n$ which encodes the network **connection strength between nodes i and j** .

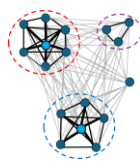
An intermediate quantity called the *co-expression similarity* s_{ij} is first defined to calculate the adjacency matrix. The default method defines the co-expression **similarity s_{ij} as the absolute value of the correlation coefficient** (Pearson, biweight or the Spearman correlation) between the profiles of nodes i and j : $s_{ij} = |\text{cor}(x_i, x_j)|$. The co-expression **similarity is transformed into the adjacency using a thresholding procedure**.

Unweighted networks: adjacency a_{ij} between gene expression profiles x_i and x_j can be defined by **hard thresholding** the co-expression similarity s_{ij} as

$$a_{ij} = \begin{cases} 1 & \text{if } s_{ij} \geq \tau; \\ 0 & \text{otherwise,} \end{cases} \quad \text{where } \tau \text{ is the "hard" threshold parameter. Thus, two genes are linked } (a_{ij} = 1) \text{ if the absolute correlation between their expression profiles exceeds the (hard) threshold } \tau.$$

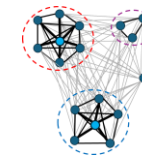
Weighted networks: the adjacency takes on continuous values between 0 and 1 and can be defined by raising the co-expression **similarity to a power**: $a_{ij} = s_{ij}^\beta$, with $\beta \geq 1$. The weighted adjacency a_{ij} between two genes is proportional to their similarity.

How to choose the threshold parameters? Scale-free topology criterion.



WGCNA

Weighted Gene Co-expression Network Analysis



How to choose the threshold parameters? Scale-free topology criterion.

A **scale-free network** is one where **most nodes** have **few connections**, and a **few nodes** (hubs) have **many connections**. Many real biological systems show this property.

WGCNA package tests a range of β values (e.g. 1 to 20), and calculates:

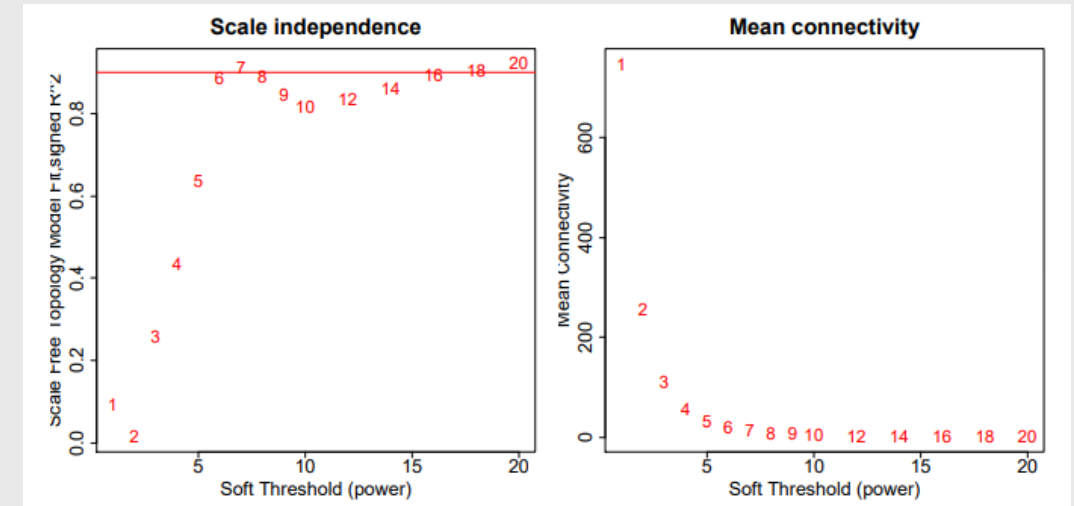
- ✓ The R^2 of the fit between node connectivity and the power-law distribution (how “scale-free” the network is)
- ✓ The mean connectivity (so the network isn’t too sparse or too dense).

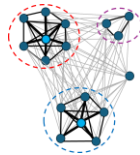
Then WGCNA plots these results:

- ✓ **R^2 vs. β** (to see where the scale-free fit becomes strong)
- ✓ **Mean connectivity vs. β** (to check network density).

The **best β** is usually the **smallest value** where:

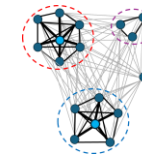
- ✓ the **scale-free fit R^2** is **high ($\geq 0.8-0.9$)**, and
- ✓ the **mean connectivity** is still reasonable.





WGCNA

Weighted Gene Co-expression Network Analysis



3. Module detection

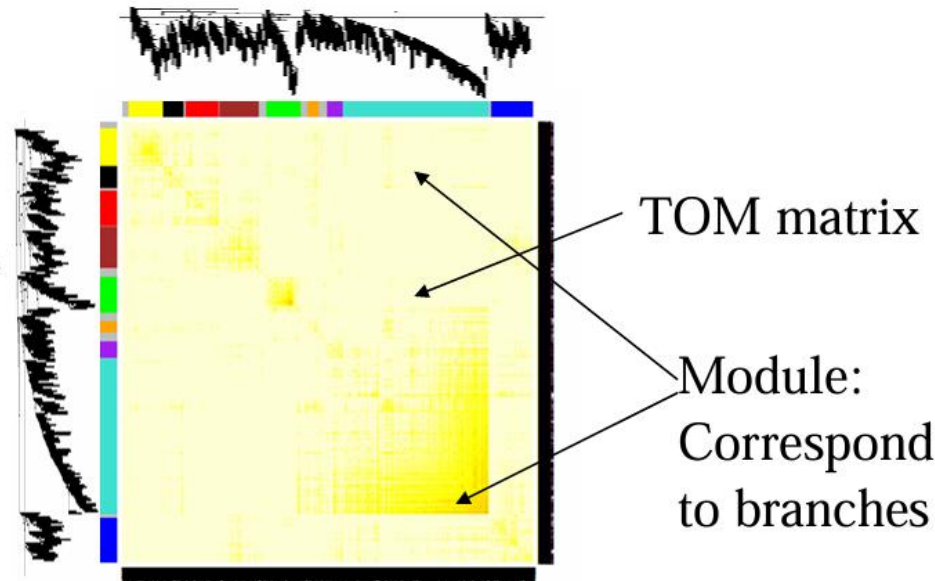
Modules are clusters of highly interconnected genes and are defined by average linkage **hierarchical clustering** using as input the **topological overlap dissimilarity** (distance) measure (the higher the value, the more different the connection patterns of two genes). $DistTOM_{ij} = 1 - TOM_{ij}$

Topological Overlap Matrix (TOM) is a measure of how similar two genes are in terms of their network connections, considering not only the direct correlation between two genes but also **how many neighbors they share** (e.g. If gene A and gene B are both connected to many of the same genes, their TOM will be high).

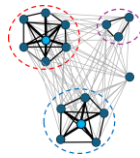
$$TOM_{ij} = \frac{\sum_u a_{iu}a_{uj} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}$$

Genes correspond to rows and columns

Hierarchical clustering dendrogram

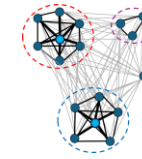


A network heatmap plot (interconnectivity plot) of a gene network together with the corresponding hierarchical clustering dendrograms and the resulting modules



WGCNA

Weighted Gene Co-expression Network Analysis



4.1. Gene significance (GS)

- ✓ It used to incorporate external information into the co-expression network.
- ✓ The higher the absolute value of GS_i , the more biologically significant is the i -th gene.
- ✓ A gene significance measure could also be defined by minus log of a p-value.

4.2. Module significance

The average absolute gene significance measure for all genes in a given module.

5. Other key parameters

Connectivity: it measures how correlated a gene is with all other network genes.

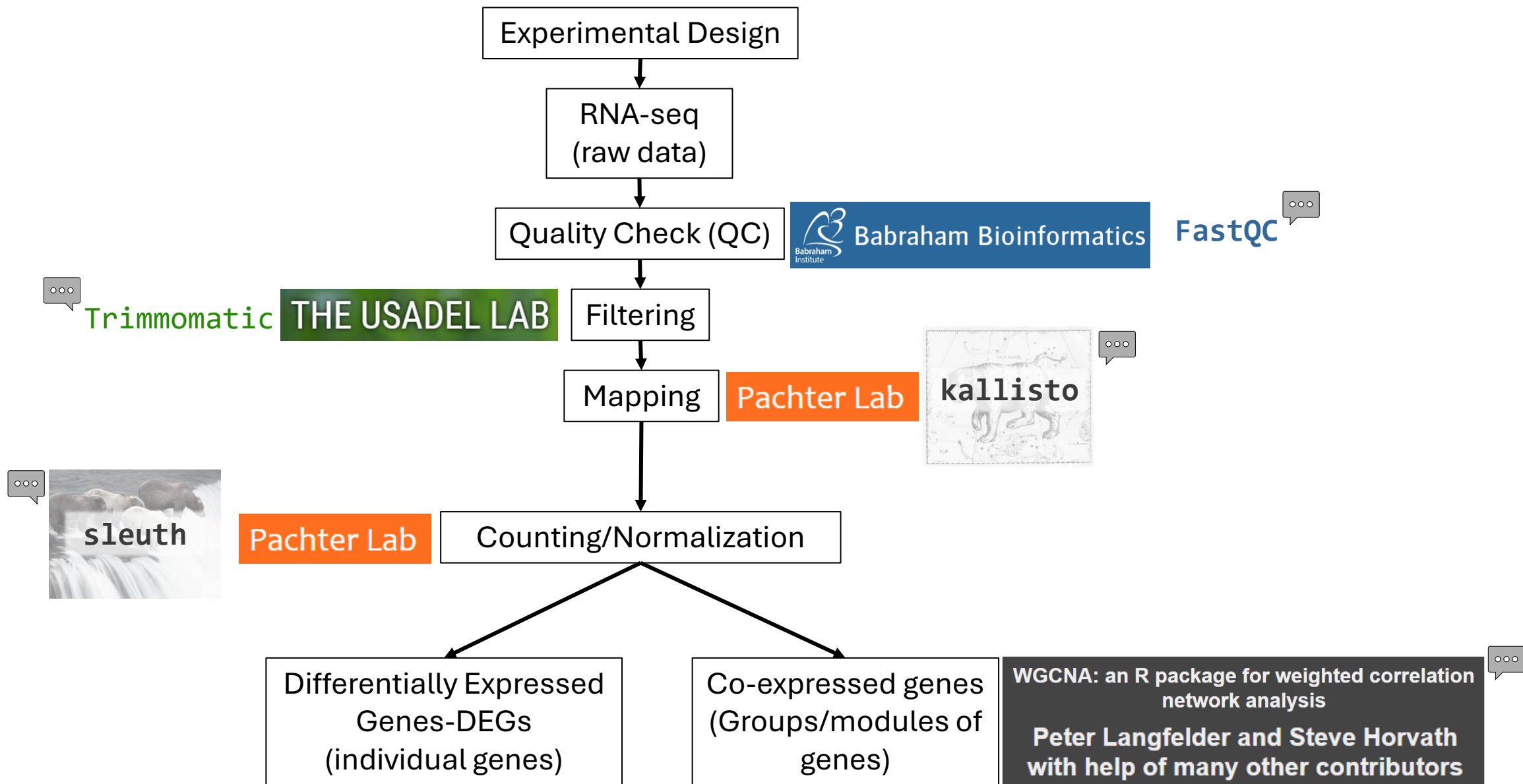
Intramodular connectivity (k_{IM}): it measures how connected, or co-expressed, a given gene is with respect to the genes of a particular module.

Module eigengene E : The first principal component of a given module (such as a representative of the gene expression profiles in a module).

Eigengene significance: The correlation coefficient between the module eigengene and a trait.

Module Membership (MM) (also known as eigengene-based connectivity k_{ME}): For each gene, a measure of module membership by correlating its gene expression profile with the module eigengene of a given module.

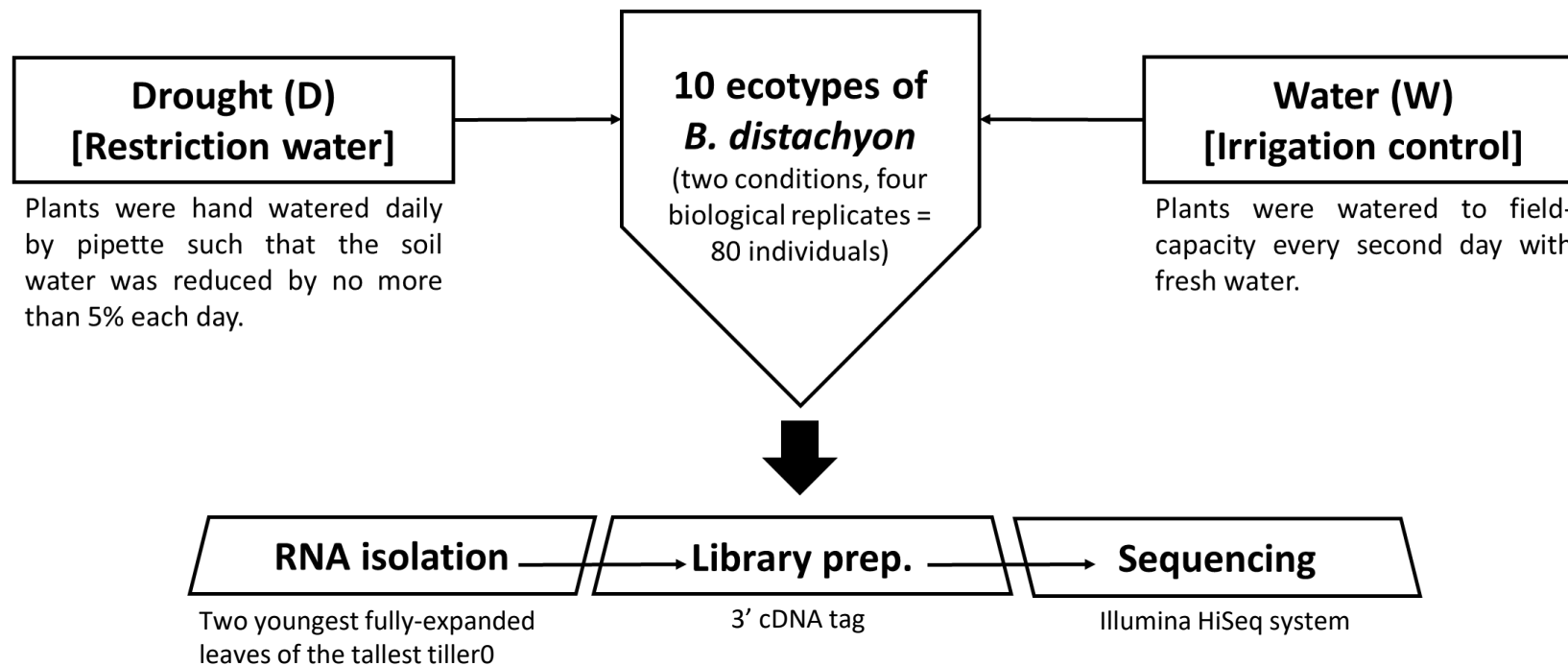
Practical Exercise



Experimental Design

Hypothesis: There are genes with similar expression profiles involved in the response to water stress.

Objective: To detect a set of co-expressed genes involved in the response to water stress and the hub genes with a key role in the regulation of the module.



Des Marais et al (2017)

Sancho et al. (2022)

Readme (Step-by-step)

- OPEN the “**Readme_Docker_Transcriptomic_class.txt**” file
 - ✓ OPEN Ubuntu (WSL)
 - ✓ In your home, create the folder "transcriptome_class" and extend permissions
 - `mkdir transcriptome_class`
 - `chmod 777 transcriptome_class`
 - ✓ Docker environment: <https://github.com/ead-csic-compbio/bioinformatics/tree/main/docker>
 - Run (in Ubuntu): `docker pull csicunam/bioinformatics_iamz:latest`
 - Bind local "transcriptome_class" folder to container folder "/data":

`docker run -it -v $HOME/transcriptome_class:/data -v /tmp/.X11-unix:/tmp/.X11-unix -e DISPLAY=$DISPLAY csicunam/bioinformatics_iamz:latest`
 - ✓ From **Windows**, the "**transcriptome_class**" folder is located in:

[\\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class](#)

[\\wsl.localhost\Ubuntu-22.04\home\rsancho\transcriptome_class](#) (Rubén Sancho laptop)

Readme (Step-by-step)

✓ Move into /data: `cd /data`

- Create symbolic link:

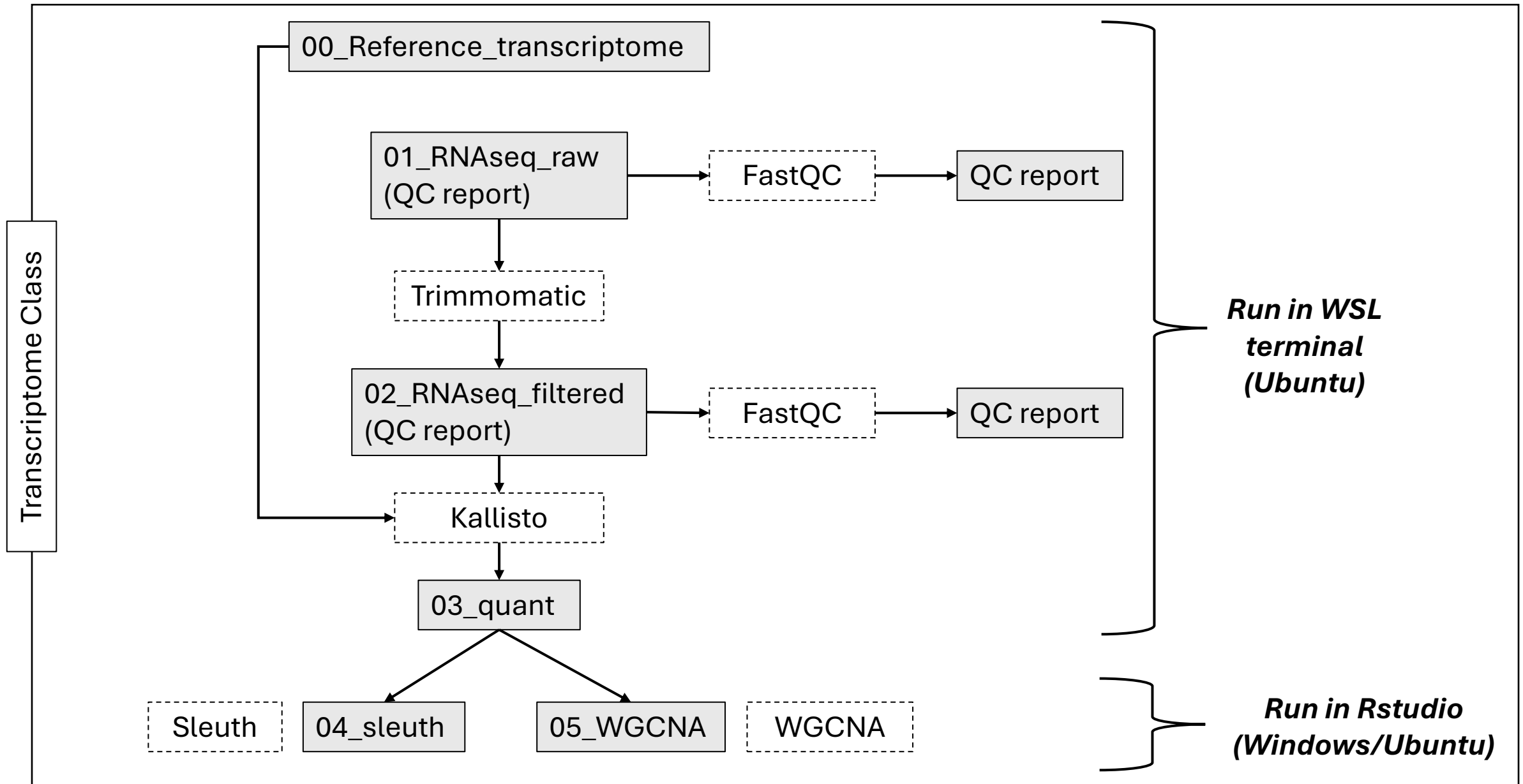
```
ln -s /home/vep/test_data/transcripts/brachy/ brachy
```

- Create report folders:

```
mkdir report report_filtered
```

Folders structure

/home/vep/test_data/transcripts/brachy/



RNA-seq

RNA-Seq library protocol is the **3' cDNA tag library prep** for sequencing on the Illumina HiSeq platform adapted from Meyer et al. (2011).

Sequencing was carried out using an Illumina HiSeq2500 platform (100 bp Single-End, SE, sequencing).

Quality Check (QC)

Task 01: Generate QC report of ABR3_W_03 sample.
Review and interpret the results obtained

Software: FastQC v0.12.1.

Command: `fastqc input_fastqc -o output_folder`

Tips:

- ✓ Remember to create the output folder.
- ✓ Check that the paths of the software, input files, and output folder are correct.

Quality Check (QC)

```
fastqc 01_RNAseq_raw/ABR3_W_03.fastq.gz \  
-o 01_RNAseq_raw/reports/
```

Open the report: \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\report\ABR3_D_03_fastqc.html

Quality Check (QC) with FastQC

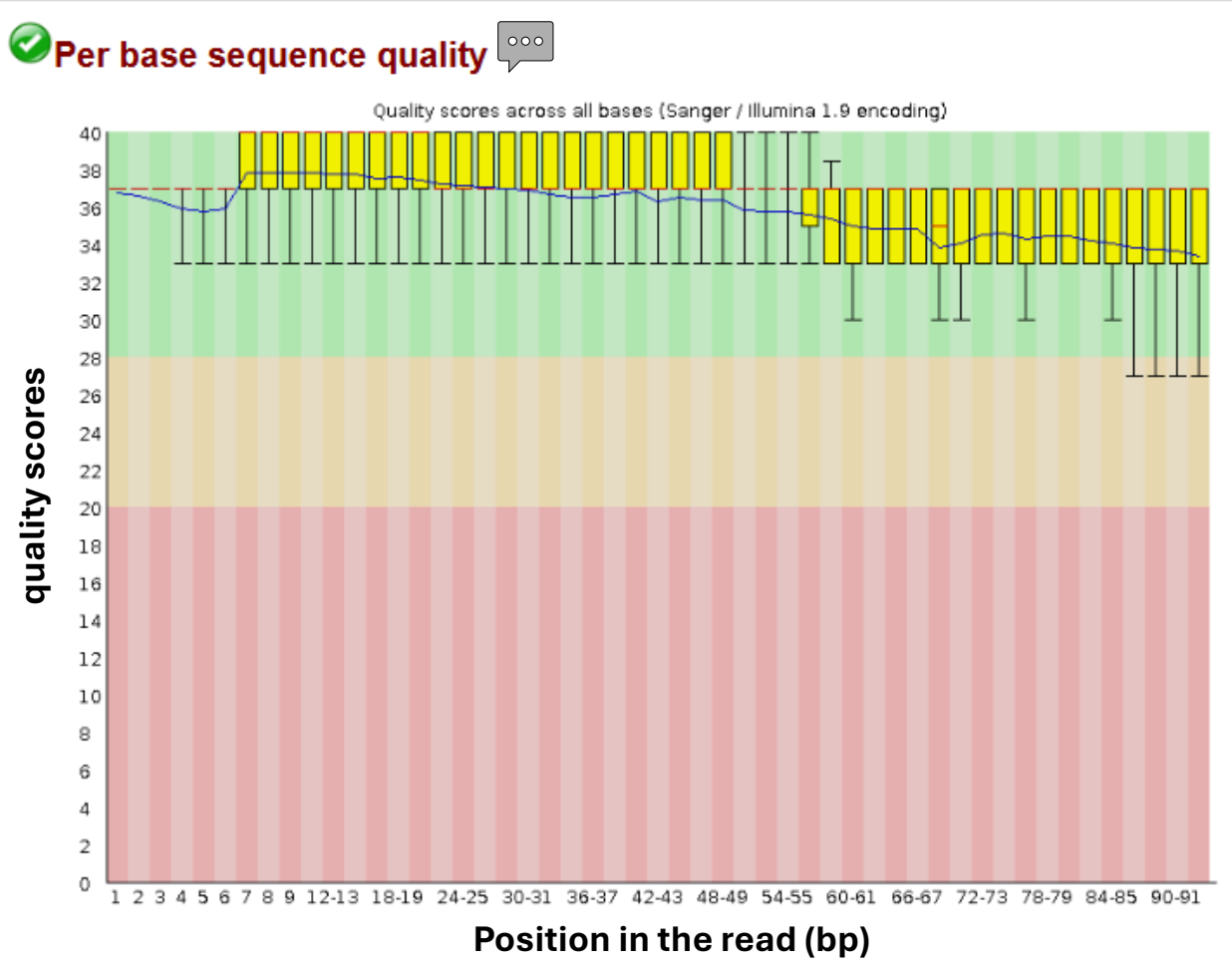
 **FastQC Report**

Summary

-  [Basic Statistics](#)
-  [Per base sequence quality](#)
-  [Per tile sequence quality](#)
-  [Per sequence quality scores](#)
-  [Per base sequence content](#)
-  [Per sequence GC content](#)
-  [Per base N content](#)
-  [Sequence Length Distribution](#)
-  [Sequence Duplication Levels](#)
-  [Overrepresented sequences](#)
-  [Adapter Content](#)

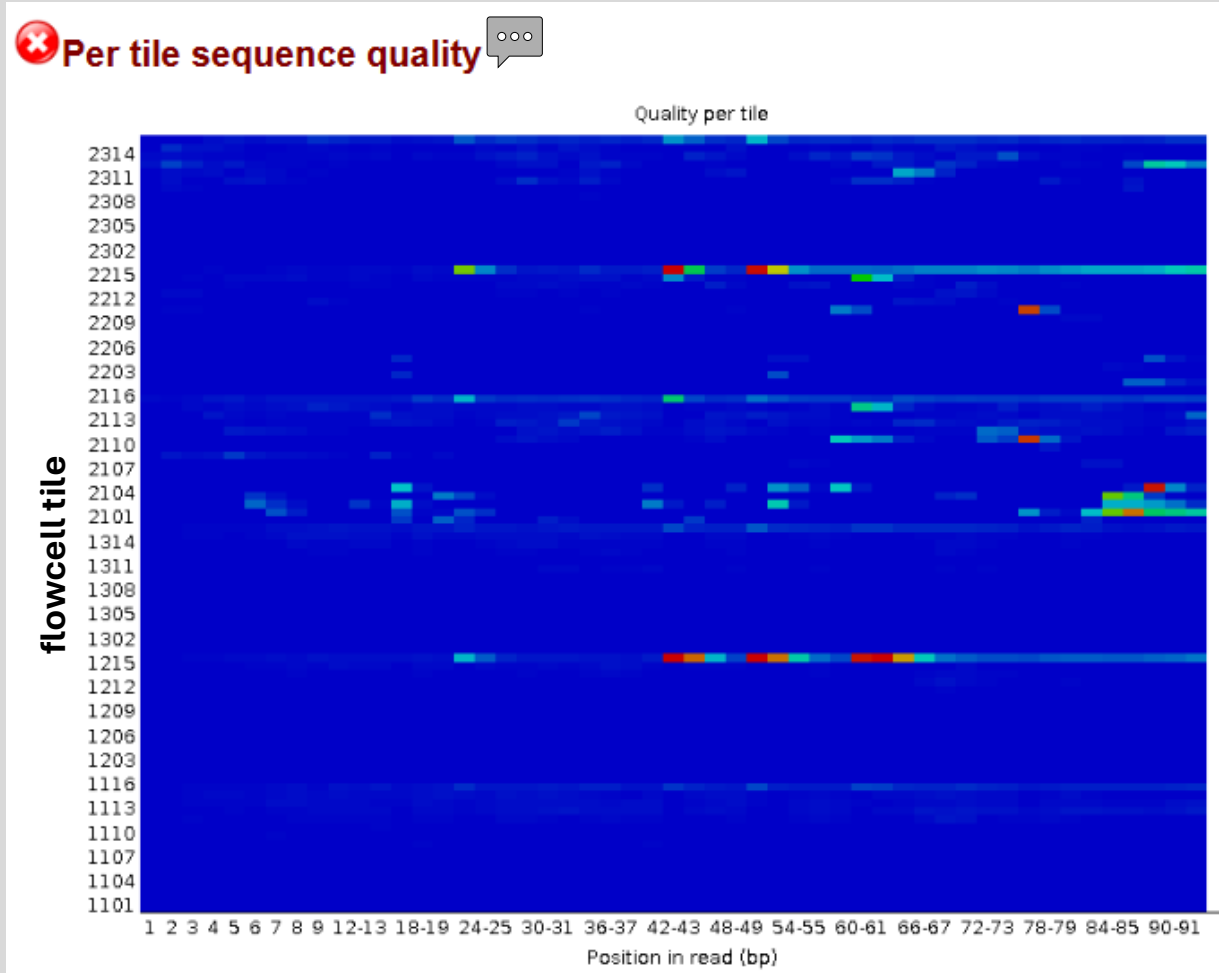
 **Basic Statistics**

Measure	Value
Filename	ABR3_W_03.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	2705509
Total Bases	248.8 Mbp
Sequences flagged as poor quality	0
Sequence length	70-93
%GC	45

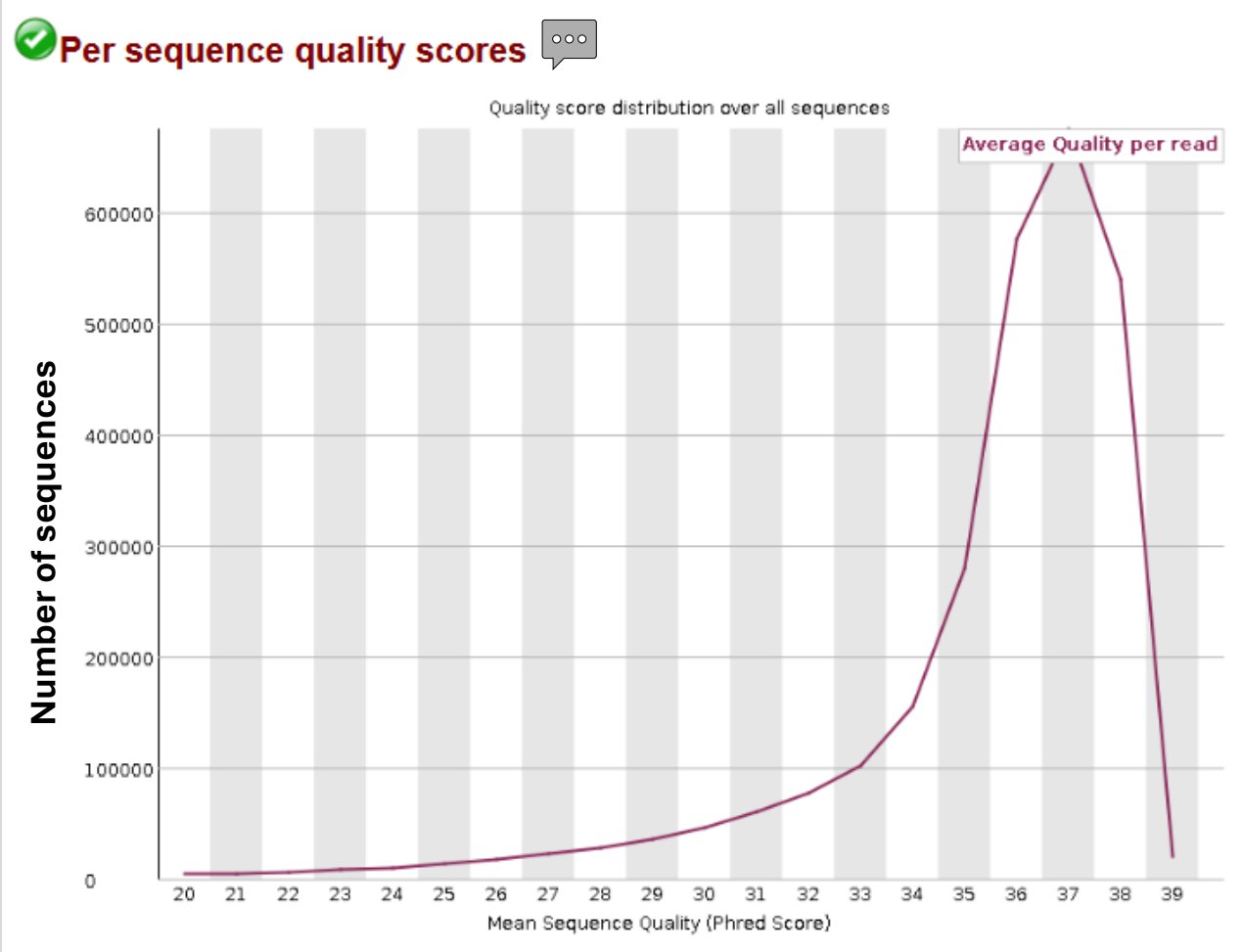


The range of quality scores across all bases at each position in the Fastq file

Quality Check (QC) with FastQC



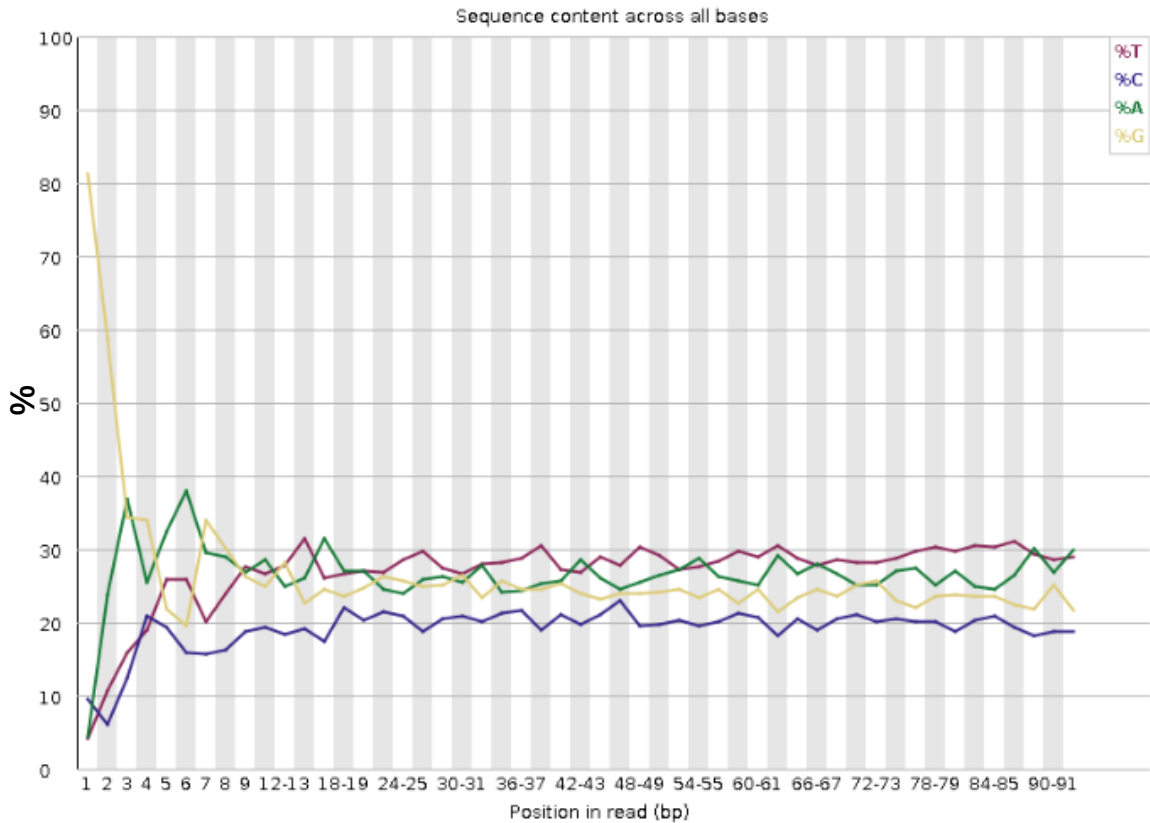
The plot shows the deviation from the average quality for each flowcell tile. Hotter colours indicate that a tile had worse qualities than other tiles for that base. Bubbles, smudges or debris in the flowcell lane.



It allows you to see if a subset of your sequences have universally low-quality values.

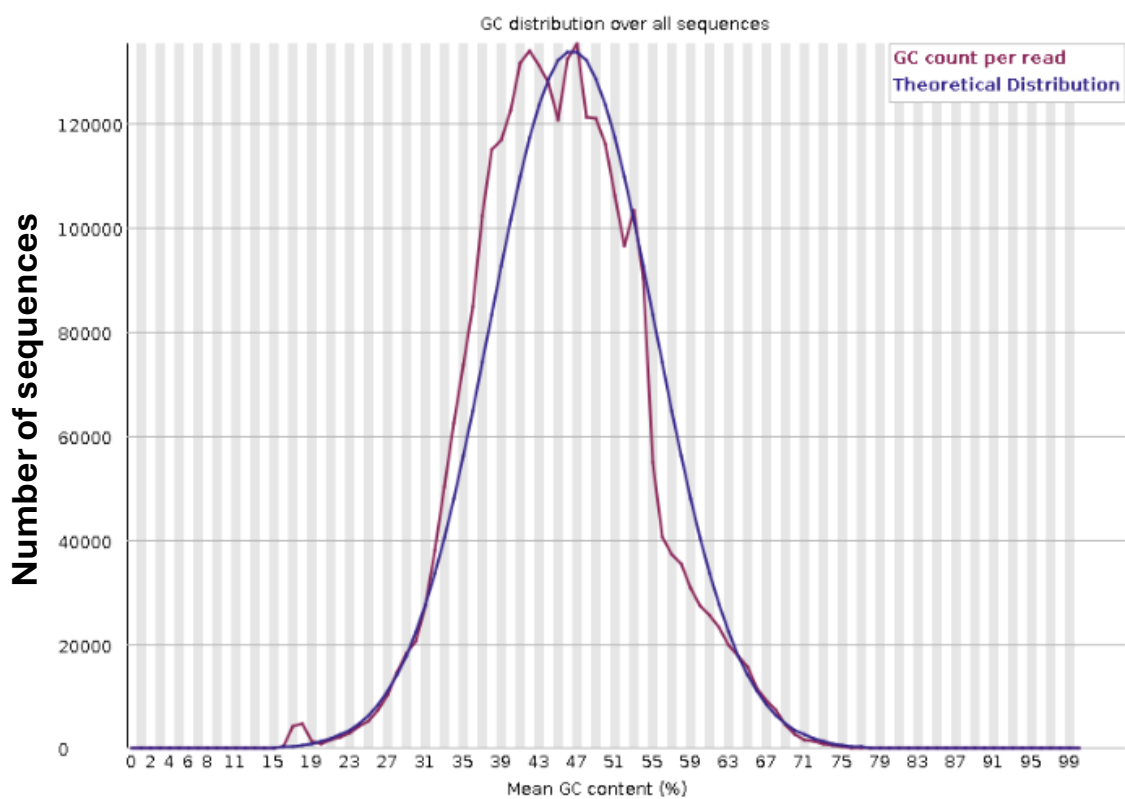
Quality Check (QC) with FastQC

✖ Per base sequence content



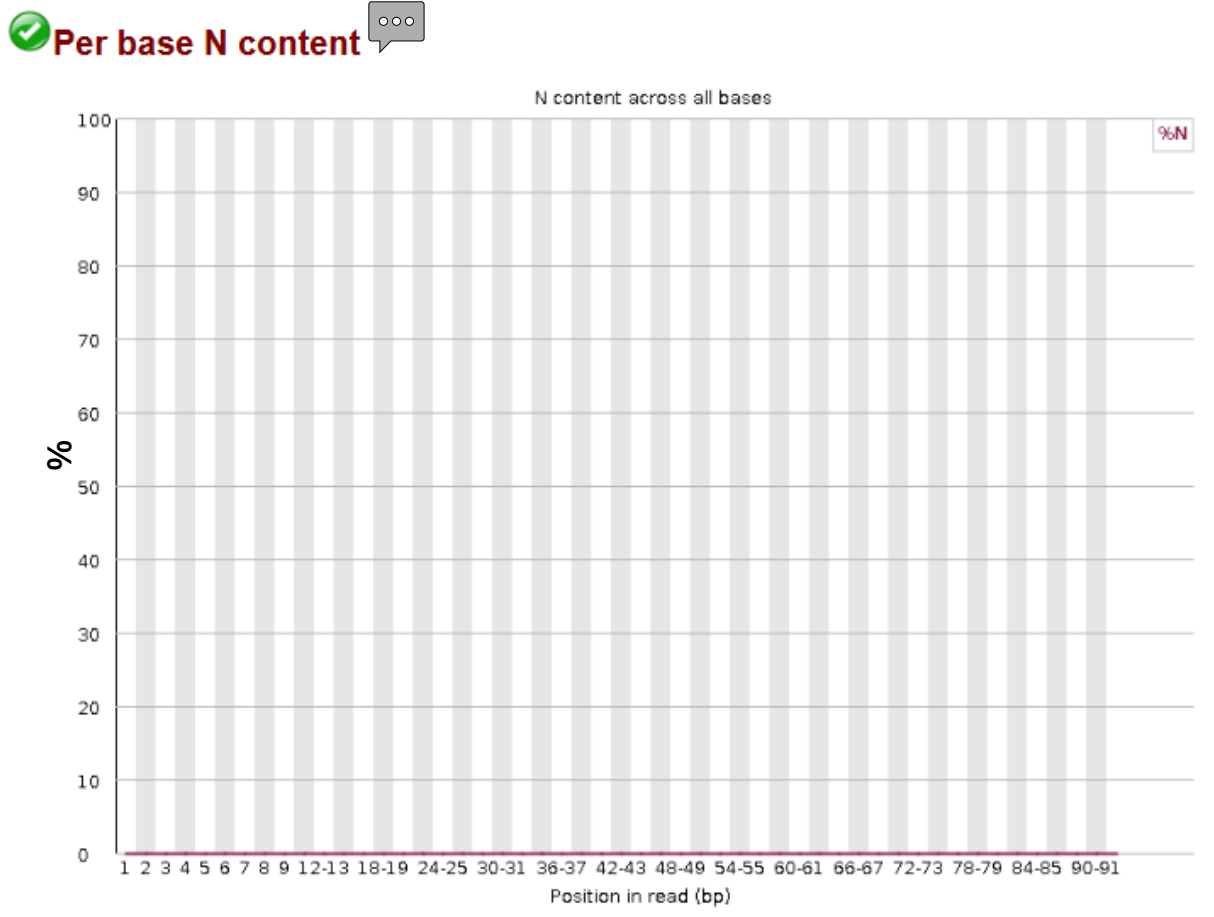
The proportion of each base position in a file for which each of the four normal DNA bases has been called. **In a random library** you would expect that there would be little to no difference between the different bases (A-T and C-G) of a sequence run. **IMPORTANT:** Some types of library will always produce biased sequence composition, normally at the start of the read. Libraries produced by priming using random hexamers (including nearly all RNA-Seq libraries) inherits an intrinsic bias in the positions at which reads start. This is not a problem which can be fixed by processing, and it doesn't seem to adversely affect the ability to measure expression.

⚠ Per sequence GC content

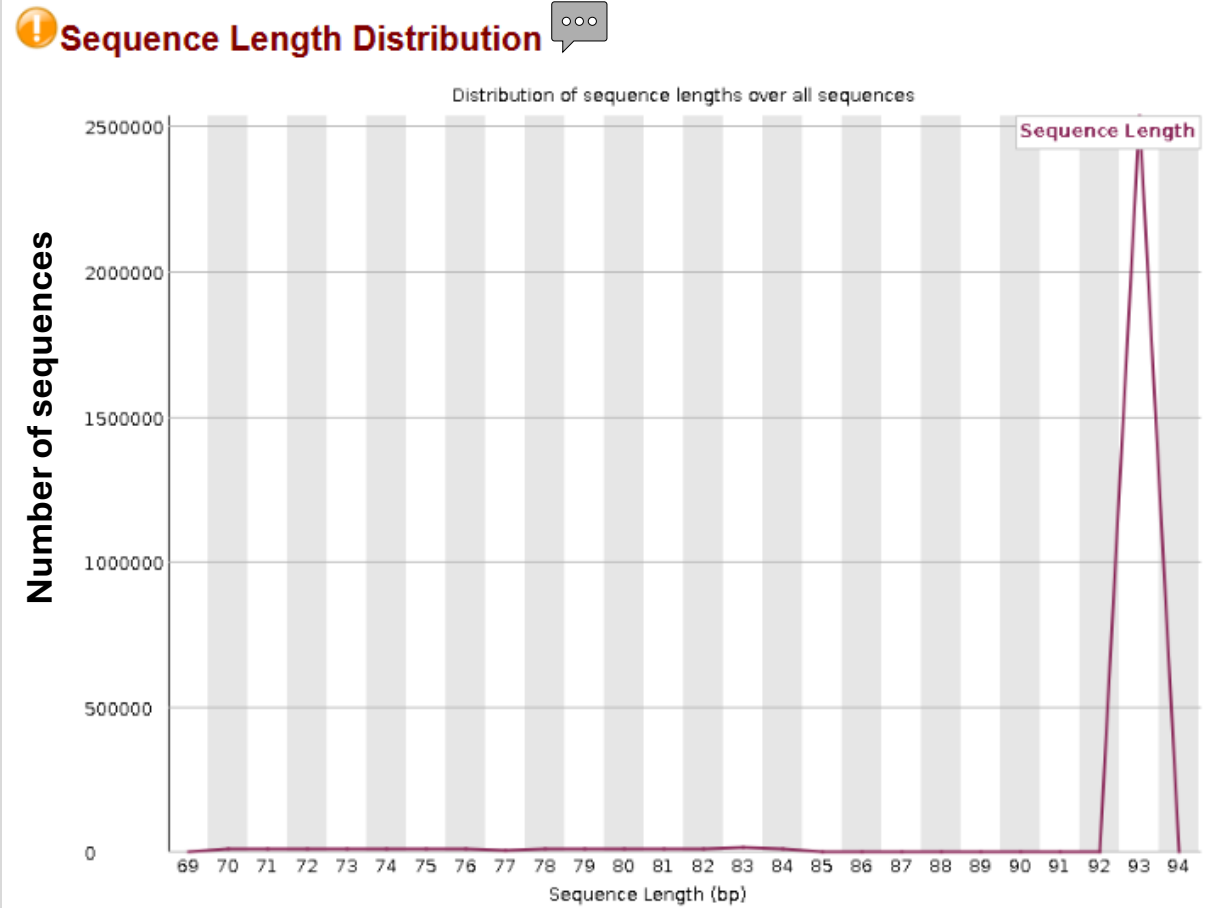


The GC content across the whole length of each sequence in a file and compares it to a modelled normal distribution of GC content. In a **normal random library**, you would expect to see a roughly normal distribution of GC content where the central peak corresponds to the overall GC content of the underlying genome.

Quality Check (QC) with FastQC

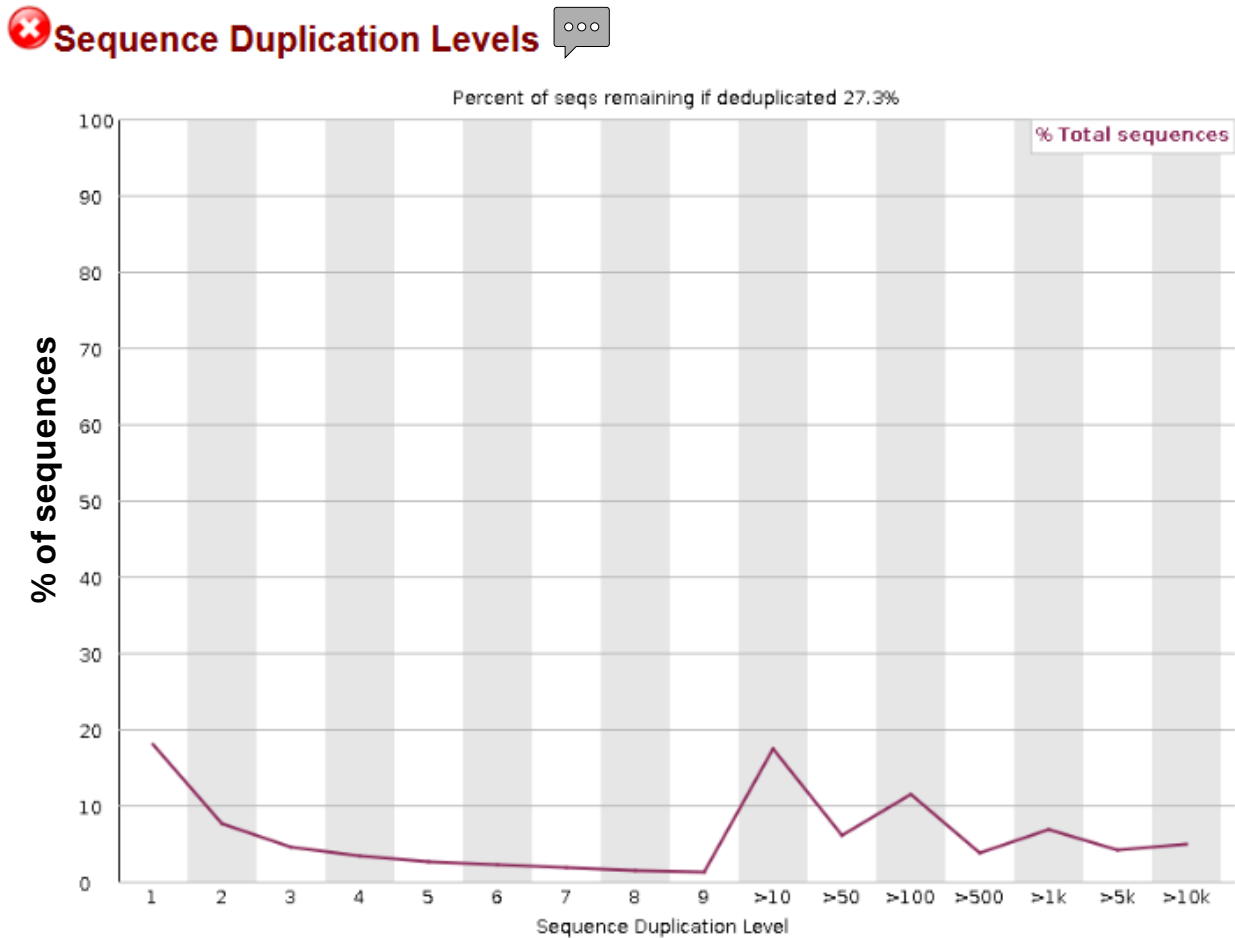


This module plots out the percentage of base calls at each position for which an N was called.



This module generates a graph showing the distribution of fragment sizes in the file which was analysed.

Quality Check (QC) with FastQC



This module counts the degree of duplication for every sequence in a library and creates a plot showing the relative number of sequences with different degrees of duplication. A high level of duplication is more likely to indicate some kind of enrichment bias.

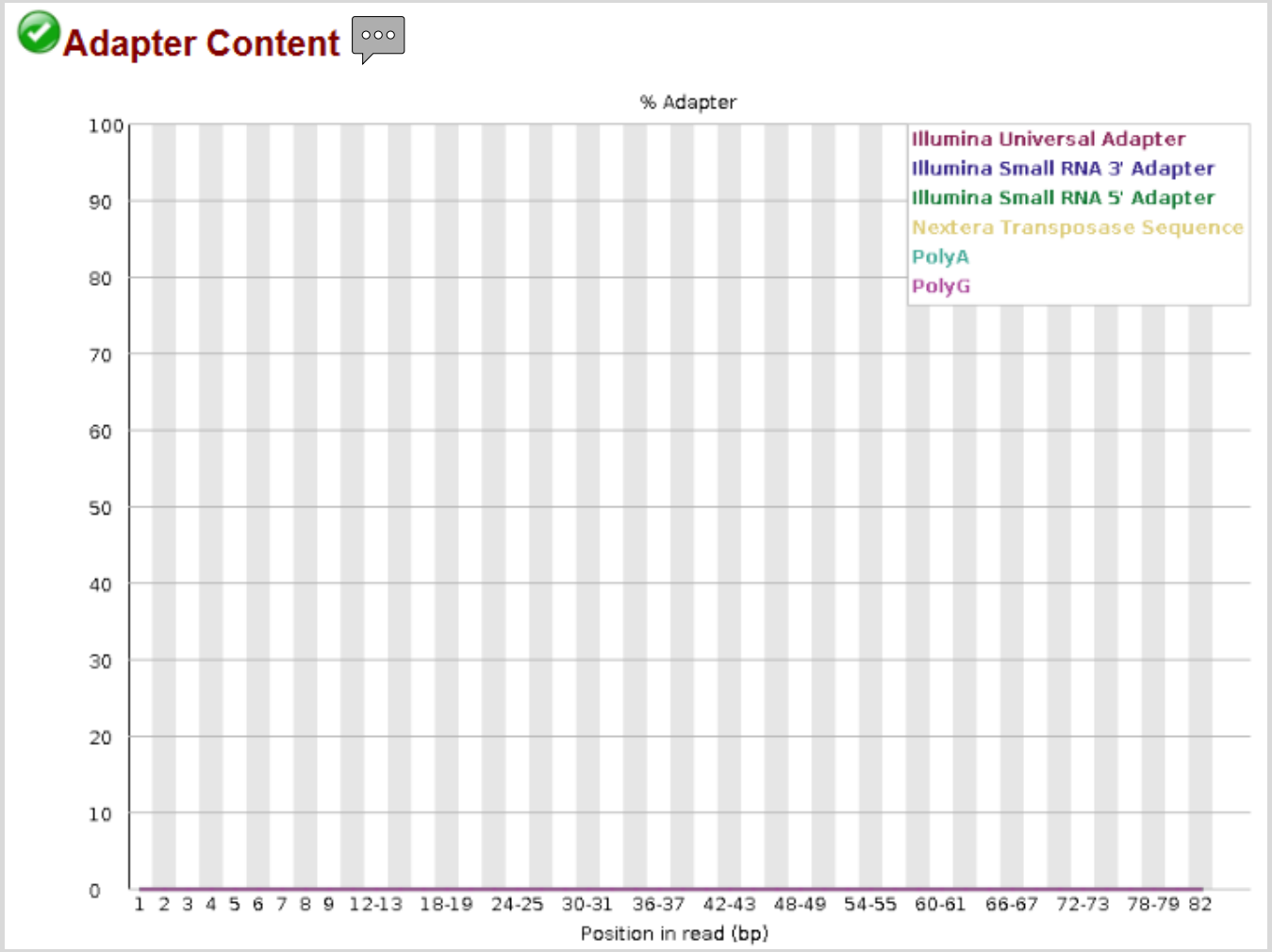
IMPORTANT: In RNA-Seq libraries, sequences from different transcripts will be present at wildly different levels in the starting population. In order to be able to observe lowly expressed transcripts it is therefore common to greatly over-sequence high expressed transcripts, and this will potentially create large set of duplicates.

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GGGGAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGT	40784	1.5074427769414185	No Hit
GGGAAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGTG	19675	0.7272199057552571	No Hit
GGGATGATTCTGTATTACAATTTGGTGGAGGAACCTTAGGACATCCTTGG	18927	0.6995726127689836	No Hit
GGACATCCTTGGGAAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTT	16776	0.6200681646226275	No Hit
GGGCGGATCTTGCTCGTGAAGTAATGAAATTATCCGAGCAGCTTGCAA	12819	0.47381102779550904	No Hit
TACCTAGGCACCCAGAGACGAGGAAGGCGTAGCAAGCGACGAAATGCTT	12385	0.4577696840040081	No Hit
GAAGGTAATGAAATTATCCGAGCAGCTTGCAAAATGGAGTCTGAACTAGC	10784	0.3985941277593236	No Hit
GGAGGAACCTTAGGACATCCTTGGGAAATGCACCTGGTGCAGCAGCTAA	9593	0.3545728363867945	No Hit
GGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGTGTACAAGCTCGTAA	9539	0.3525769088182667	No Hit
GGGGATGATTCTGTATTACAATTTGGTGGAGGAACCTTAGGACATCCTTG	8802	0.32533619367002664	No Hit
GGATGATTCTGTATTACAATTTGGTGGAGGAACCTTAGGACATCCTTGGG	8521	0.3149499779893543	No Hit
GGGAGAGCAATACAAGCGTTGTGCTGCTAGGCGAAGCGGTTGAGTGCCGC	8453	0.3124365877178749	No Hit
GGTAATGAAATTATCCGAGCAGCTTGCAAAATGGAGTCTGAACTAGCCGC	8362	0.30907308014868917	No Hit
GGAGTATCCTTGATATATAAATATATAGATAAATAGATTTAAATGGAAAA	8070	0.2982802866299835	No Hit
GTATGGAAGGCGATCAAAATTCGAGTTCGCGCCGGTAGATACTATCGATTA	7873	0.29099884716702107	No Hit
GTATTTAGCCTTGCAAGGTGGTCTTGCTGATTACACGGGATTCCACGT	6178	0.2283488984882327	No Hit
GGAGTCTGAACTAGCCGAGCTTGTGAAGTATGGAAGGCGATCAAAATTC	5746	0.21238147794001055	No Hit

This module lists all of the sequence which make up more than 0.1% of the total. For each overrepresented sequence the program will look for matches in a database of common contaminants and will report the best hit it finds. This module will often be triggered when used to analyse small RNA libraries where sequences are not subjected to random fragmentation, and the same sequence may naturally be present in a significant proportion of the library.

Quality Check (QC) with FastQC



It checks if your library contains a significant amount of adapters in order to be able to assess whether you need to adapter trim or not. The plot itself shows a cumulative percentage count of the proportion of your library which has seen each of the adapter sequences at each position

Filtering with Trimmomatic

Task 02: Filter the sequences of ABR3_W_03 sample. Select the most appropriate parameters based on the QC report to improve their quality.

Software: Trimmomatic v.0.39

Command: `java -jar trimmomatic-0.39.jar SE -threads 4 -phred33 \`
`Input_file Output_file\`
`ILLUMINACLIP:path/adapter_file:x:xx:xx \`
`LEADING:xx TRAILING:xx \`
`SLIDINGWINDOW:xx:xx \`
`MINLEN:xx`

Tips:

- ✓ Check that the paths of the software, input files, and output folder are correct.
- ✓ *Phred+33 is listed as Illumina 1.9/Sanger (see QC report)*
- ✓ *Adapter file TruSeq3-SE.fa for Illumina HiSeq2500*
- ✓ Check Trimmomatic website: <http://www.usadellab.org/cms/?page=trimmomatic>

Filtering with Trimmomatic

```
java -jar trimmomatic-0.39.jar \  
SE \  
-threads 4 \  
-phred33 \  
01_RNAseq_raw/ABR3_W_03.fastq.gz \  
02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz \  
ILLUMINACLIP:software/Trimmomatic-0.39/adapters/TruSeq3-SE.fa:2:30:10 \  
LEADING:3 \  
TRAILING:3 \  
SLIDINGWINDOW:4:15 \  
MINLEN:36
```

**Quality Check (QC)
after filtering step**

Task 03: Generate a new QC report of the filtered sequences and compare the reports before and after filtering. What parameters have changed?

Software: FastQC v.0.12.1

Comparing QC reports before and after filtering step

```
fastqc 02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz -o 02_RNAseq_filtered/reports/
```

Before ...

 **Basic Statistics**

Measure	Value
Filename	ABR3_W_03.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	2705509
Total Bases	248.8 Mbp
Sequences flagged as poor quality	0
Sequence length	70-93
%GC	45

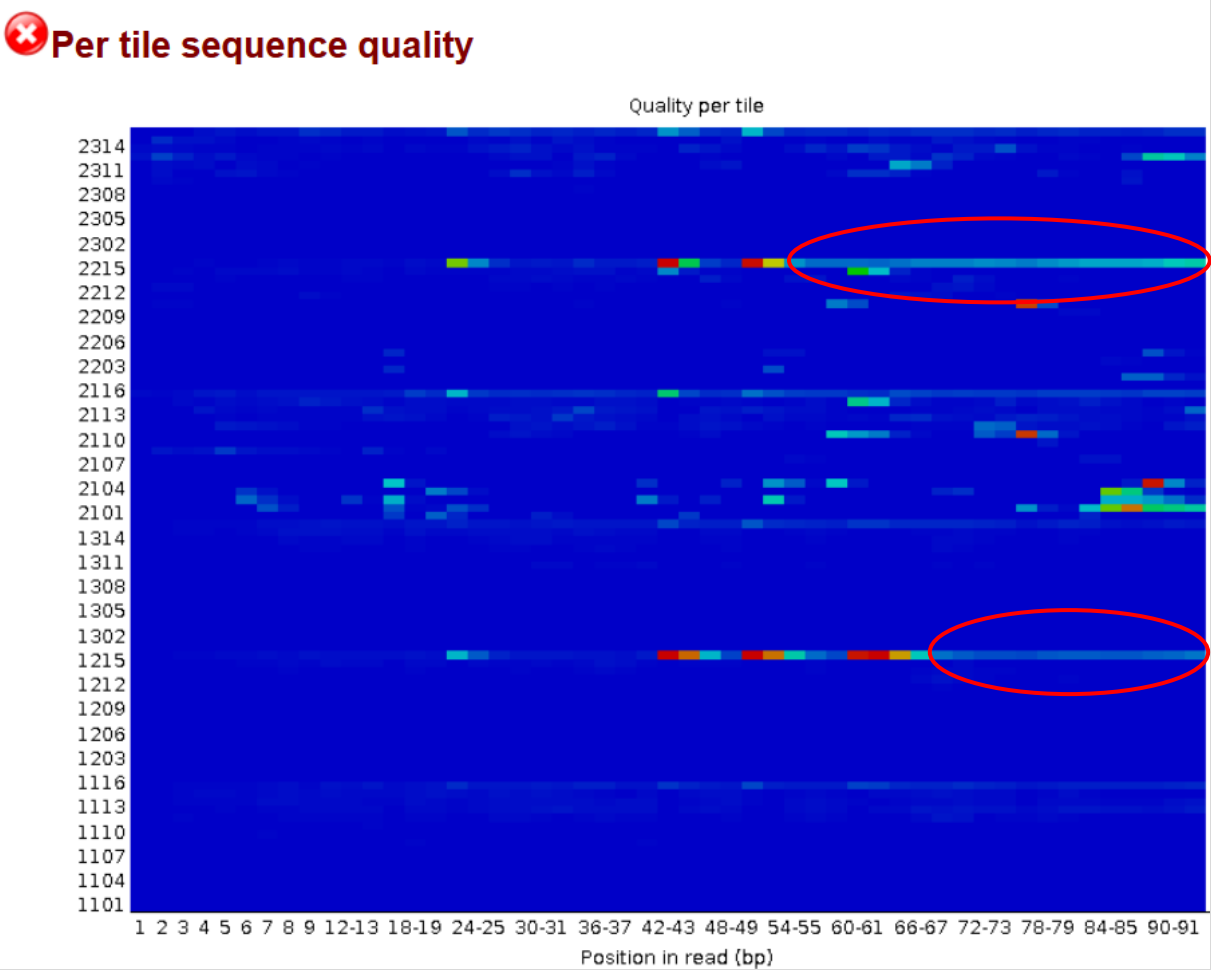
After ...

 **Basic Statistics**

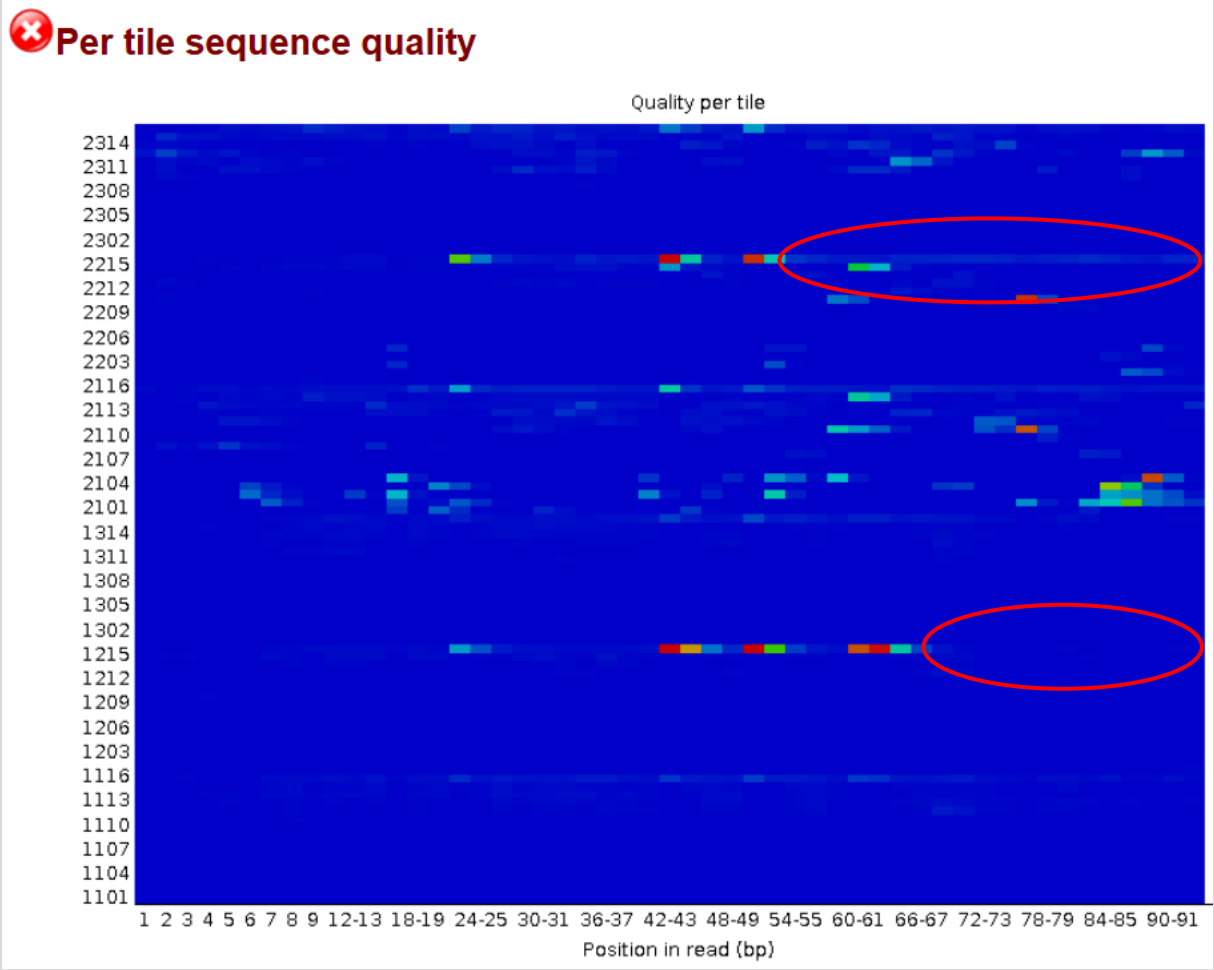
Measure	Value
Filename	ABR3_W_03_trimmed.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	2666619
Total Bases	241.2 Mbp
Sequences flagged as poor quality	0
Sequence length	36-93
%GC	45

Comparing QC reports before and after filtering step

Before ...



After ...



Mapping with kallisto

Task 04: Perform pseudo-alignments of the reads from sample ABR3_W_03 against the reference transcriptome (Bdistachyon_314_v3.1.transcript.TagSeq500b.fa.gz) and indicate the expression of the Bradi3g51200 gene in est_counts and TPM.

Software: kallisto 0.48

Index reference

```
kallisto index \  
-i ref.output.idx ref.input.fa.gz
```

Quantification

```
kallisto quant \  
-i ref.output.idx \  
-o output_folder/output_subfolder_sample \  
-b 100 \           # Number of bootstrap samples*  
--single \         # Quantify single-end reads  
-l 100 \           # Estimated average fragment length**  
-s 20 \            # Estimated standard deviation of fragment length***  
<(zcat input_compressed.fastq.gz)
```

Mapping with kallisto

- * Number of bootstrap samples
- ** Estimated average fragment length
- *** Estimated standard deviation of fragment length

Number of bootstrap samples (-b): Because many reads can map to multiple transcripts, kallisto uses a probabilistic model to assign them. To measure how uncertain these assignments are, kallisto performs bootstrap resampling — it repeatedly resamples reads (with replacement) and re-estimates transcript abundances. Typical bootstrapping value: 100

Estimated average fragment length (-l): In the case of single-end reads, the -l option must be used to specify the average fragment length. Typical Illumina libraries produce fragment lengths ranging from 180–200 bp but it's best to determine this from a library quantification with an instrument such as an Agilent Bioanalyzer. For paired-end reads, the average fragment length can be directly estimated from the reads, and the program will do so if -l is not used (this is the preferred run mode).

Estimated standard deviation of fragment length (-s): This value describes how much variation there is in the length of the sequenced cDNA fragments. Library kits or sequencing facilities often report the average fragment size and its variation, e.g. “200 ± 20 bp” → -l 200 -s 20. Typical ‘s’ value: 20

Mapping with kallisto

Pseudoalignments of RNA-Seq data were carried out using as references 500 bp from the 3' tails of the *B. distachyon*_314_version 3.1 **transcriptome** (IBI, 2010; <https://phytozome.jgi.doe.gov/>).

We applied estimated average fragment lengths of 100 bp and standard deviations of fragment length of 20.

Index reference

```
kallisto index \  
-i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx \  
00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.fa.gz
```

Quantification

```
kallisto quant \  
-i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx \  
-o 03_quant/ABR3_W_03 \  
-b 100 \           # Number of bootstrap samples  
--single \         # Quantify single-end reads  
-l 100 \           # Estimated average fragment length  
-s 20 \            # Estimated standard deviation of fragment length  
<(zcat 02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz)
```

Mapping with kallisto

Output files:

- **abundance.h5**: a HDF5 binary file containing run info, abundance estimates, bootstrap estimates, and transcript length information length. This file can be read in by sleuth.
- **abundance.tsv**: a plaintext file of the abundance estimates. It does not contain bootstrap estimates. The first line contains a header for each column, including estimated counts, TPM, effective length.
- **run_info.json**: a json file containing information about the run.

abundance.tsv

target_id	length	eff_length	est_counts	tpm
Bradi0012s00100.1	500	401	34.4331 18.6309	
Bradi0012s00201.1	500	401	0.000217899	0.0001179
Bradi0012s00201.2	500	401	0.000217899	0.0001179
Bradi0012s00201.3	500	401	0.000217899	0.0001179
Bradi0012s00201.4	500	401	2.2899 1.23901	
Bradi0012s00201.5	500	401	0 0	
Bradi0014s00100.1	500	401	0 0	
Bradi0135s00100.1	500	401	0 0	
Bradi0180s00100.1	500	401	0 0	
Bradi1g00200.1	500	401	0 0	
Bradi1g00210.3	500	401	0 0	
Bradi1g00213.1	500	401	0 0	
Bradi1g00215.1	500	401	0 0	
Bradi1g00217.1	381	282	0 0	
Bradi1g00219.1	500	401	1 0.541076	
Bradi1g00222.1	324	225	13.952 13.4541	

run_info.json

```
{
  "n_targets": 52972,
  "n_bootstraps": 100,
  "n_processed": 2666619,
  "n_pseudoaligned": 1828214,
  "n_unique": 1262059,
  "p_pseudoaligned": 68.6,
  "p_unique": 47.3,
  "kallisto_version": "0.48.0",
  "index_version": 10,
  "start_time": "Thu Oct 30 15:01:55 2025",
  "call": "kallisto quant -i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx -o 03_quant/ABR3_W_03 -b 100 --single -l 100 -s 20 /dev/fd/63"
}
```

Mapping with kallisto

Gene expression for Bradi3g51200 (sample ABR3_W_03):

✓ **est_counts = 60**

✓ **TPM = 32.4646**

Readme (Sleuth and WGCNA)

✓ Into /data (Ubuntu WSL)

- `cp -r /home/vep/test_data/transcripts/brachy/05_WGCNA/ .`
- `cp -r /home/vep/test_data/transcripts/brachy/04_Sleuth/ .`
- Extent permissions: `chmod -R 777 04_Sleuth 05_WGCNA`
- Copy Rstudio scripts:
 - Copy Rstudio script "Practical_Sleuth.rmd" in
`\\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\04_Sleuth`
 - Copy Rstudio script "Practical_WGCNA_W_dataset.rmd" in
`\\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA`
 - Copy Rstudio script "Practical_WGCNA_D_dataset.rmd" in
`\\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA`

Data exploration and DEGs with Sleuth

Task 05: Run the Sleuth pipeline to explore your data and DEGs.

Software: `sleuth v.0.30.2`

Pipeline in Rstudio: `Practical_Sleuth.Rmd`

In \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\04_Sleuth

Data exploration and DEGs with Sleuth

Task 05.1: Use `sleuth_live(so)` to explore the boxplots of the Bradi1g06560.1 isoform/transcript expression (in TPMs) with respect to the experimental conditions (drought and irrigation).

Sleuth_live(so): analyses → transcript view

transcript view

Boxplots of transcript abundances showing technical variation in each sample.

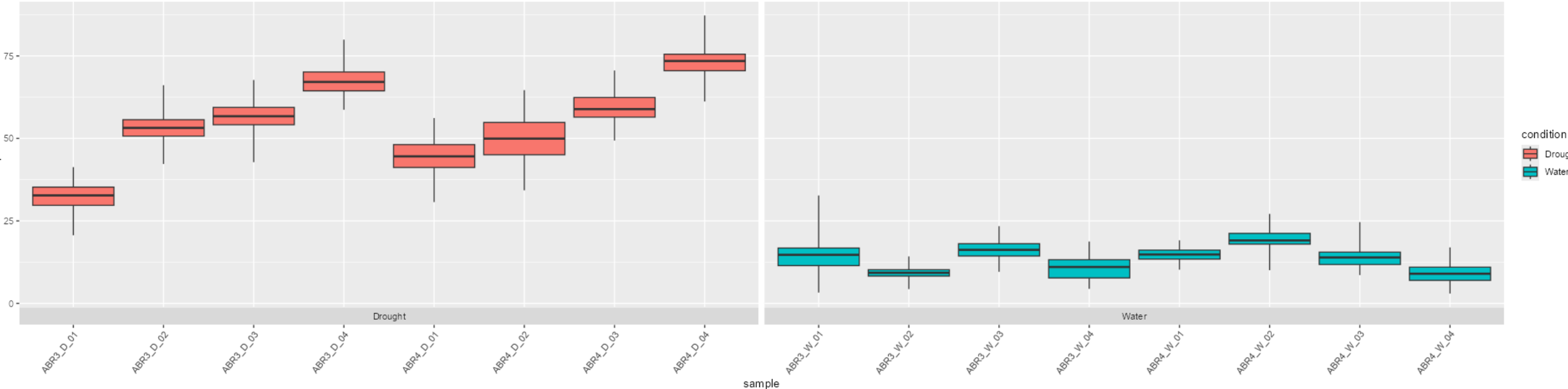
transcript:

color by:

units:

view

Bradi1g06560.1



Data exploration and DEGs with Sleuth

Task 05.2: Use `sleuth_live(so)` to explore the heatmap of the following isoforms: Bradi1g06560.1 Bradi2g54810.2 Bradi4g08240.1 Bradi4g31150.1. Use as covariate “condition” and as units TPMs.

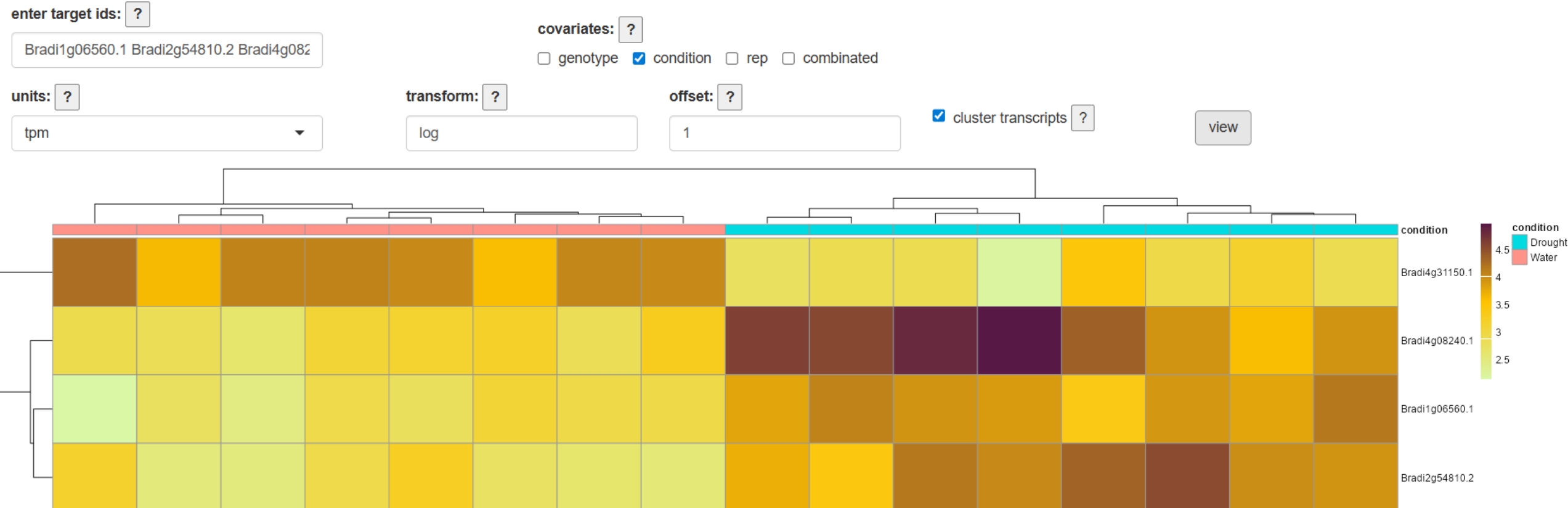
Sleuth_live(so): analyses → heatmap

sleuth overview analyses maps summaries diagnostics settings

heat map

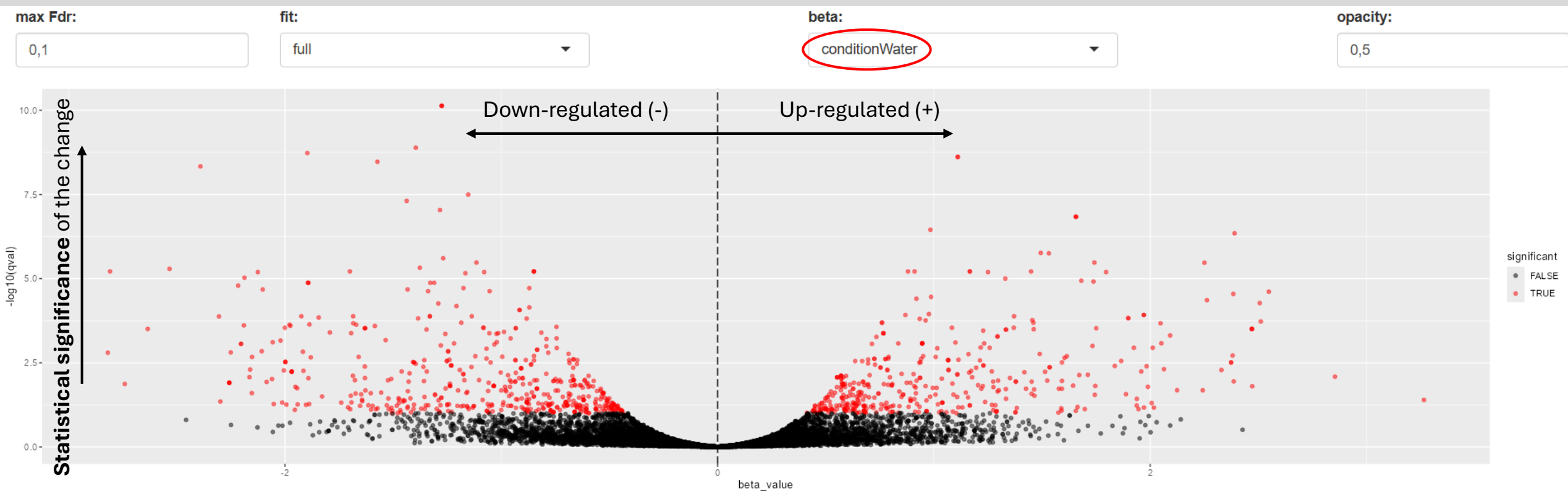
Plot of select abundances in a clustered heat map using the R [hclust function](#). The heat map iteratively groups together transcripts with similar expression patterns, highlighting similarities among the transcripts themselves. For a more in-depth discussion of hierarchical clustering, see [10.1007/BF02289588](#).

Enter space-separated values (The ten transcripts with the lowest q-values are already entered. Click view to see them!).



Data exploration and DEGs with Sleuth

Sleuth_live(so): analyses → volcano plot



X-axis (log2 fold change) This represents the **magnitude of change in gene expression** between two conditions.

Data exploration and DEGs with Sleuth

Sleuth_live(so): summaries → densities

sleuth overview analyses maps summaries diagnostics settings

distribution of abundances

Distributions of abundances of individual samples or groupings by covariates.

grouping:

condition ▼

units: ?

tpm ▼

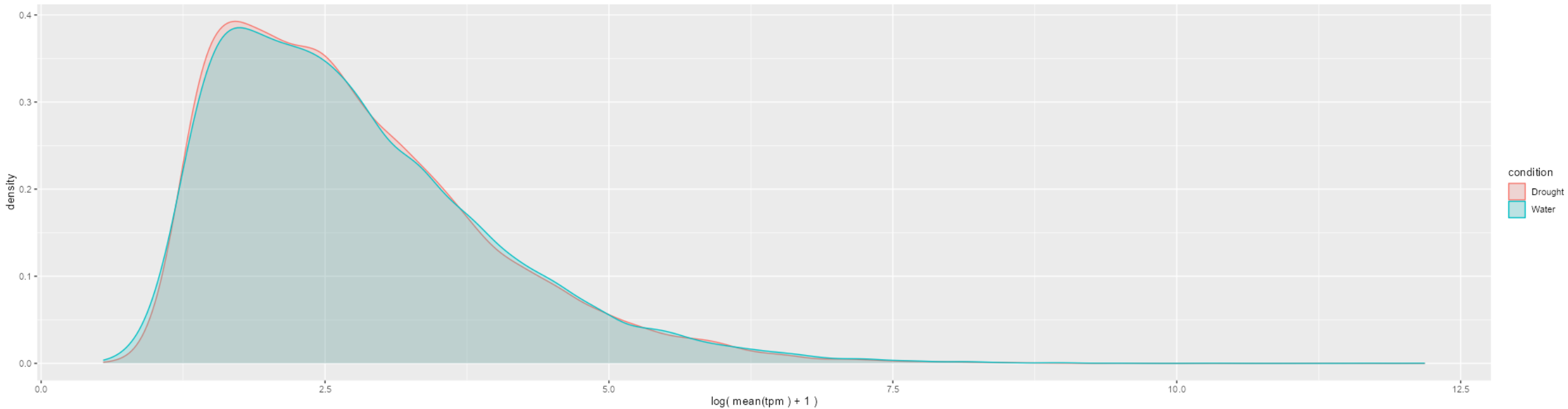
☒ filter ?

transform: ?

log

offset: ?

1

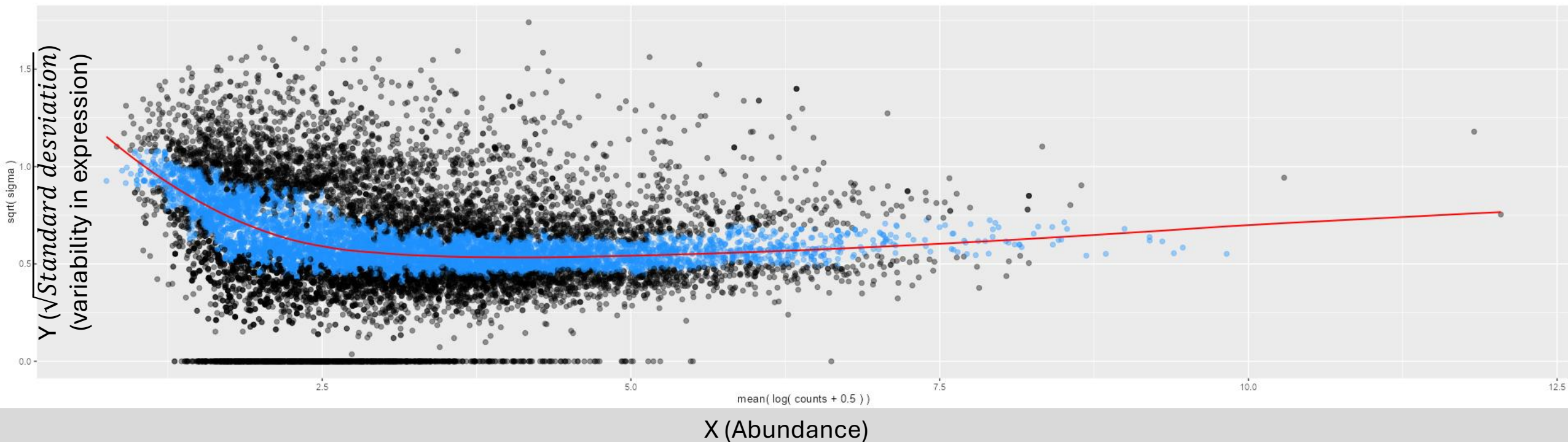


Data exploration and DEGs with Sleuth

Sleuth_live(so): diagnostics → mean-variance plot

mean-variance plot

Plot of abundance versus square root of standard deviation which is used for shrinkage estimation. The blue dots are in the interquartile range and the red curve is the fit used by sleuth. Any points at $y = 0$ have inferential variance greater than observed total raw variance.



Data exploration and DEGs with Sleuth

Sleuth_live(so): diagnostics → Q-Q plot

Q-Q plot

Select the test and view the appropriate quantile-quantile plot. Points that are color red are considered "significant" at the selected fdr-level.

max Fdr:

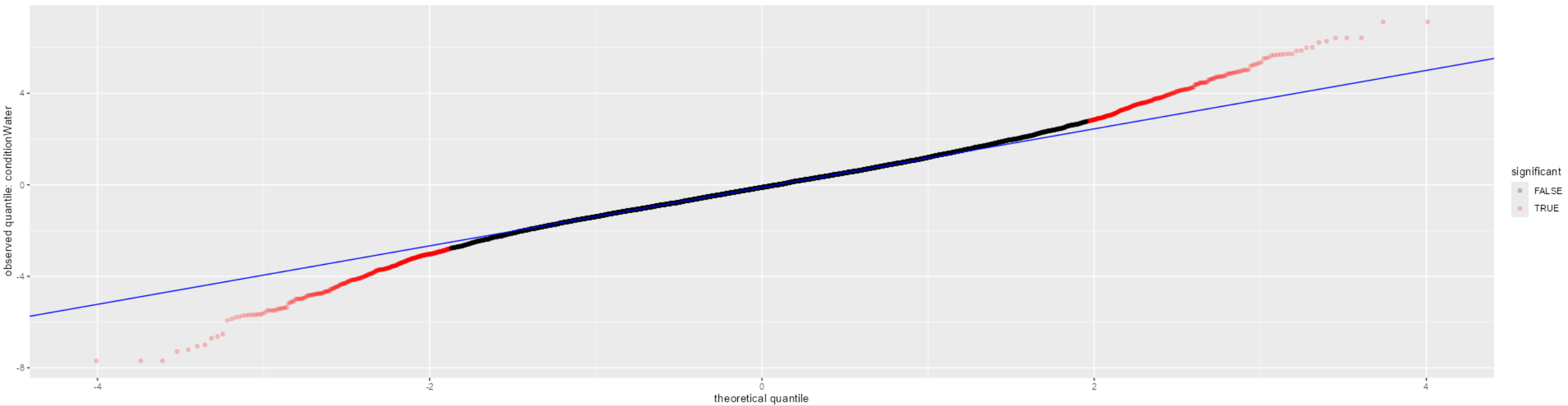
0,1

fit:

full

beta:

conditionWater



Data exploration and DEGs with Sleuth

- ✓ **Sleuth** is a powerful R package designed to analyze gene expression values calculated by the **Kallisto** software (pseudo-aligner).
- ✓ **Sleuth** allows for **exploring** and **visualizing** the data, which is essential for this type of analysis.
- ✓ **Sleuth** also allows for the identification of **differentially expressed genes (DEGs)**.

Co-expression Network with WGCNA

Task 05: Run the WGCNA pipeline to construct and explore the co-expression network.

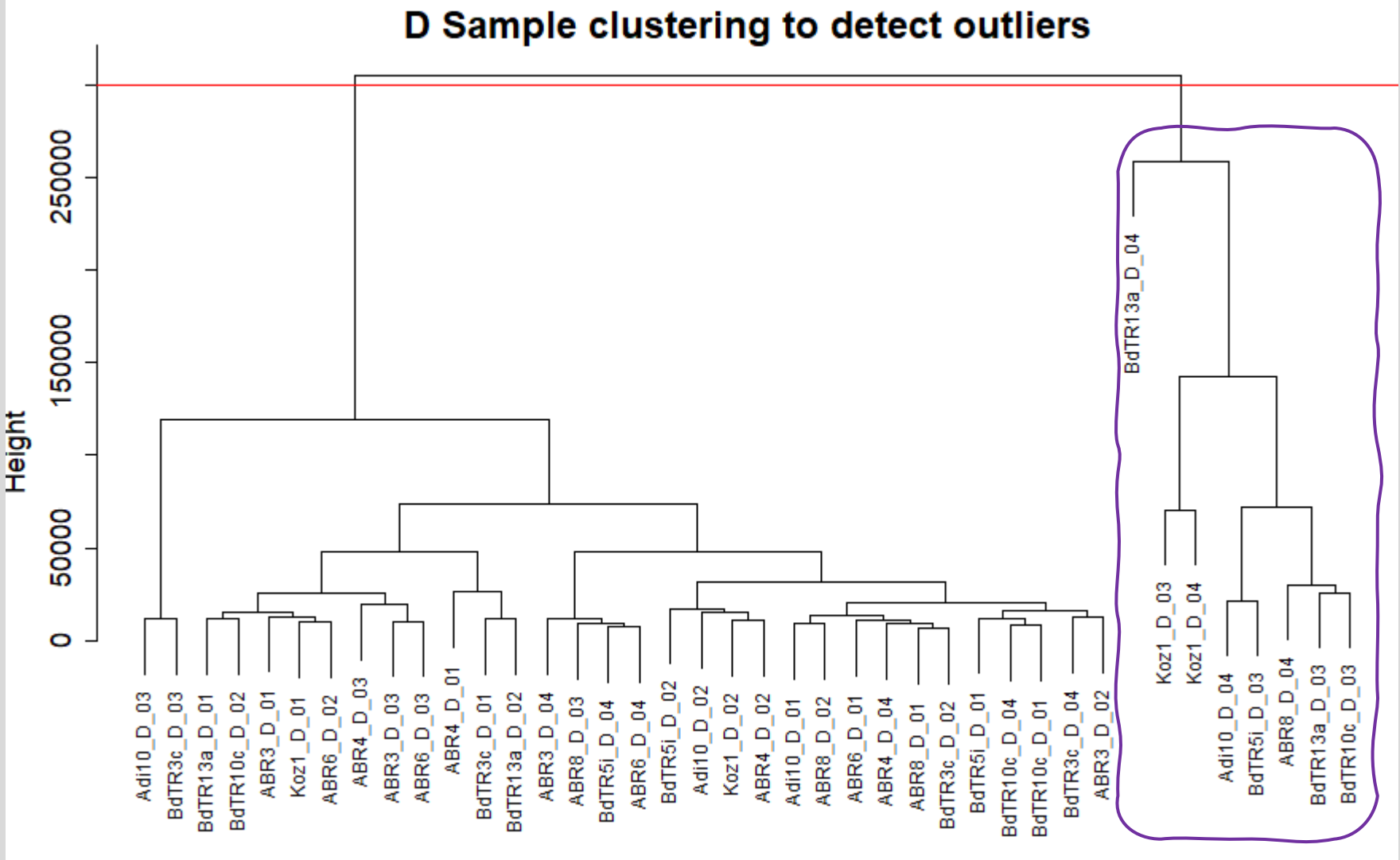
Software: WGCNA v.1.73

Pipeline in Rstudio: Practical_WGCNA.Rmd

In \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA

Co-expression Network with WGCNA

Outliers Detection

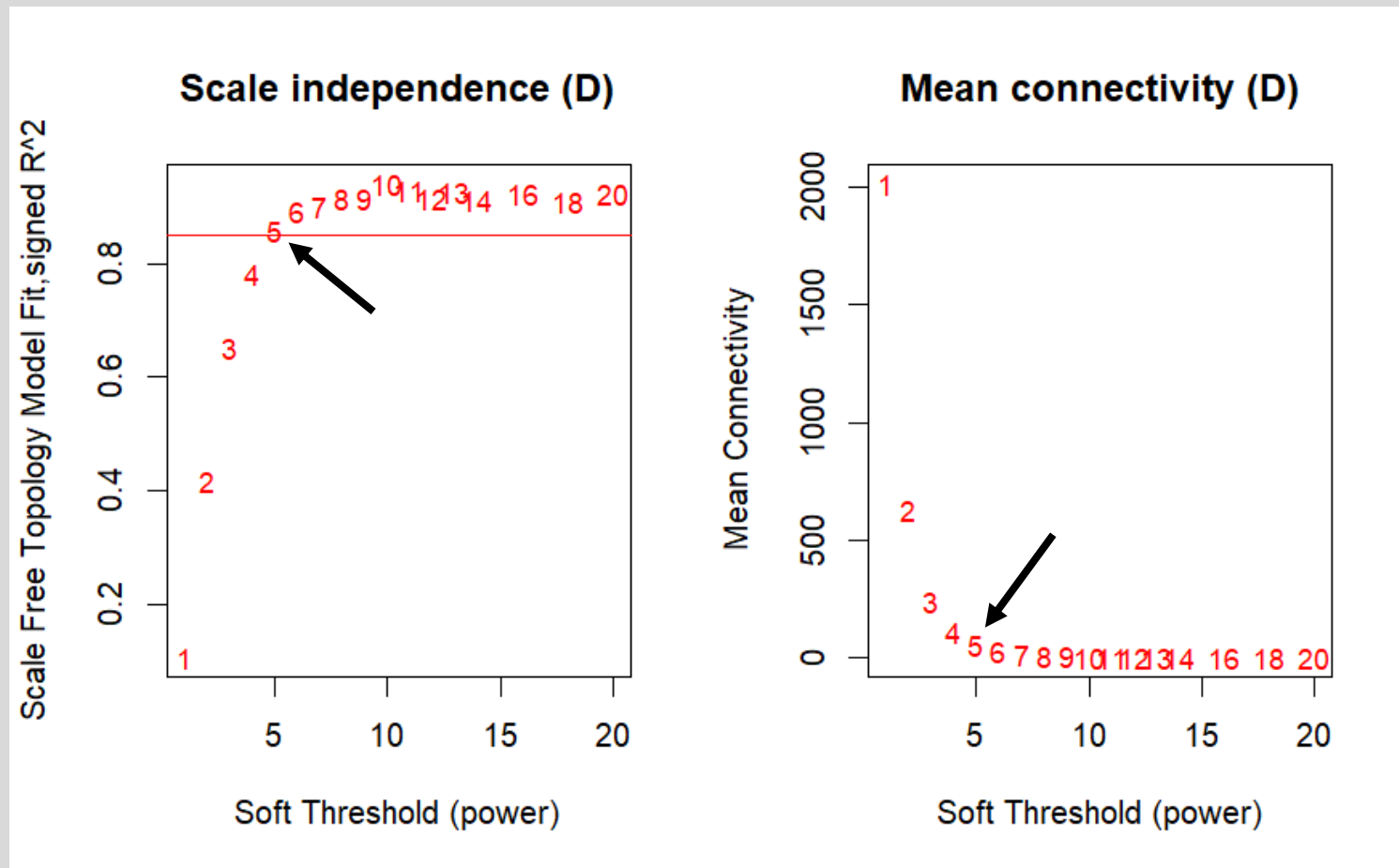


Co-expression Network with WGCNA

Soft-thresholding powers selection for network construction

$$a_{ij} = s_{ij}^{\beta}$$

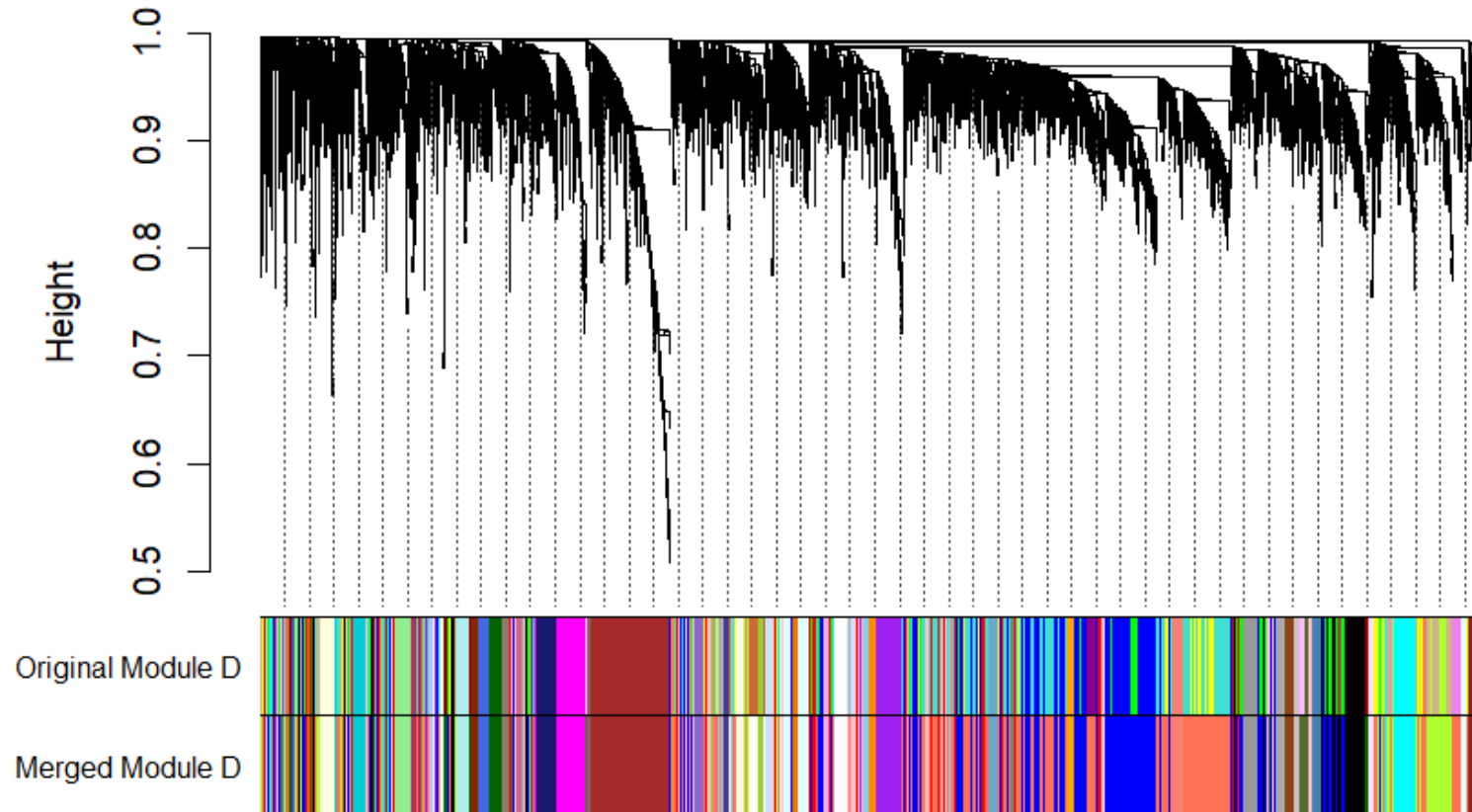
Power <dbl>	SFT.R.sq <dbl>	mean.k. <dbl>
1	0.106	2010.000
2	0.417	627.000
3	0.652	243.000
4	0.782	109.000
5	0.858	54.800
6	0.891	29.900
7	0.902	17.500
8	0.914	10.900
9	0.916	7.230
10	0.941	5.030



Co-expression Network with WGCNA

Network construction and Module detection

Dendrogram and modules for original and merged modules in D datas

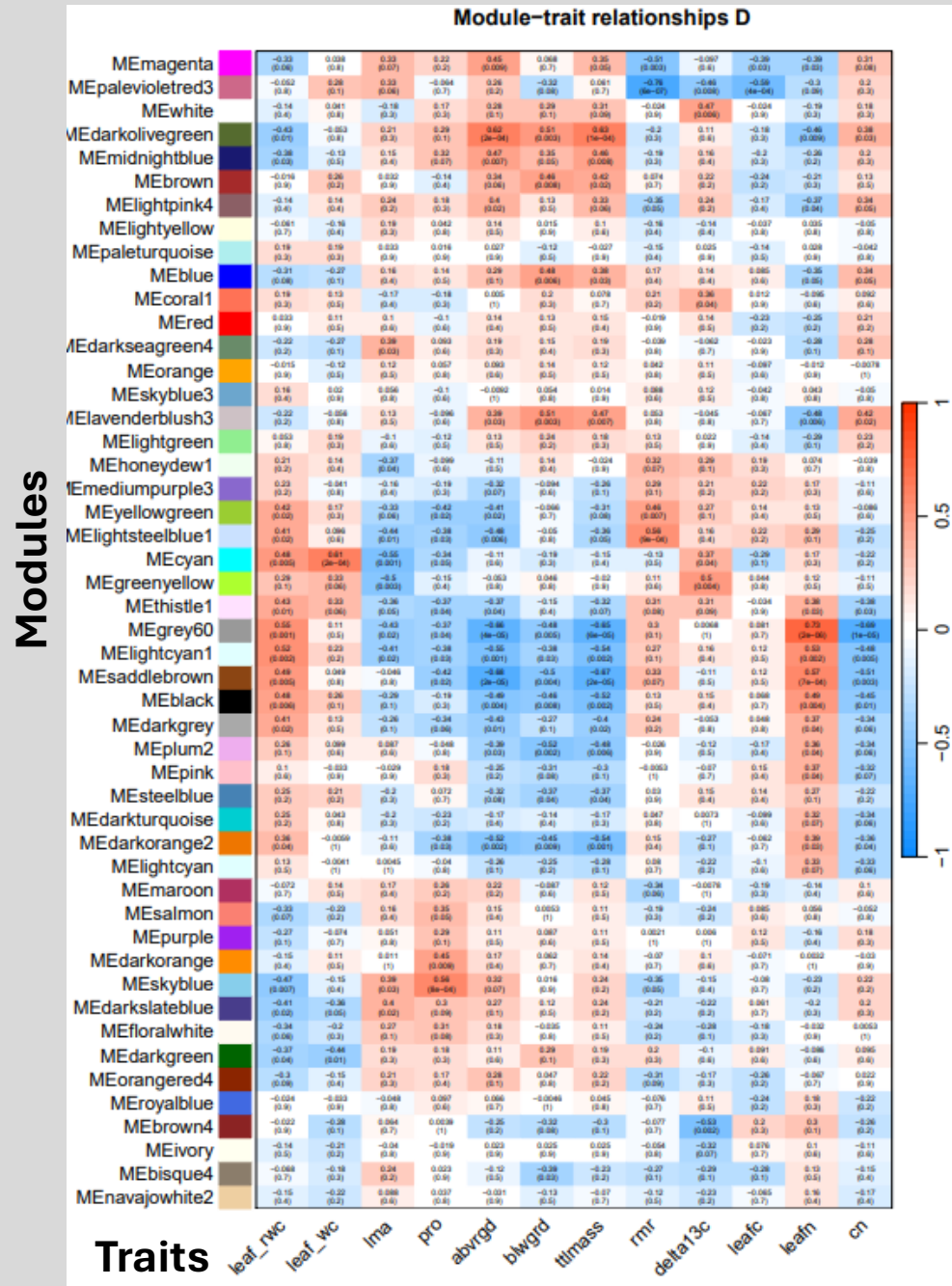


The **gray** module represents unclustered genes (without an assigned module).

Note: In this exercise, very few genes appear in "gray" because I filtered them beforehand to reduce the database size and speed up computation.

Co-expression Network with WGCNA

Module-trait associations



Co-expression Network with WGCNA

Task 06. Perform a Statistical overrepresentation test (<https://pantherdb.org/>) to determine if the genes of the magenta module of the drought network are significantly involved in any biological process according to the "GO"

(The format of the ID gene must be converted: `sed 's/Bradi/BRADI_/g' ID.txt | sed -E 's/\. [0-9]/v3/g' | sort | uniq | tr '\n' ' ')`

BRADI_1g01507v3 BRADI_1g03590v3 BRADI_1g03610v3 BRADI_1g04440v3 BRADI_1g07237v3 BRADI_1g08410v3 BRADI_1g08870v3 BRADI_1g09950v3 BRADI_1g11630v3
BRADI_1g11960v3 BRADI_1g13232v3 BRADI_1g14940v3 BRADI_1g15030v3 BRADI_1g16190v3 BRADI_1g16620v3 BRADI_1g20360v3 BRADI_1g20620v3 BRADI_1g20930v3
BRADI_1g26260v3 BRADI_1g27490v3 BRADI_1g29040v3 BRADI_1g30247v3 BRADI_1g32330v3 BRADI_1g32660v3 BRADI_1g32990v3 BRADI_1g33300v3 BRADI_1g35742v3
BRADI_1g36290v3 BRADI_1g37080v3 BRADI_1g37175v3 BRADI_1g37750v3 BRADI_1g38760v3 BRADI_1g42730v3 BRADI_1g44047v3 BRADI_1g44230v3 BRADI_1g45080v3
BRADI_1g45160v3 BRADI_1g45200v3 BRADI_1g46530v3 BRADI_1g47767v3 BRADI_1g48010v3 BRADI_1g50370v3 BRADI_1g51610v3 BRADI_1g53320v3 BRADI_1g53850v3
BRADI_1g55500v3 BRADI_1g55620v3 BRADI_1g61040v3 BRADI_1g61170v3 BRADI_1g64850v3 BRADI_1g65480v3 BRADI_1g66800v3 BRADI_1g67040v3 BRADI_1g67290v3
BRADI_1g68720v3 BRADI_1g69400v3 BRADI_1g70700v3 BRADI_1g72740v3 BRADI_1g72950v3 BRADI_1g73120v3 BRADI_1g74300v3 BRADI_1g76950v3 BRADI_1g77637v3
BRADI_2g00831v3 BRADI_2g02110v3 BRADI_2g02400v3 BRADI_2g05640v3 BRADI_2g07040v3 BRADI_2g12830v3 BRADI_2g13110v3 BRADI_2g13870v3
BRADI_2g13870v30 BRADI_2g13870v33 BRADI_2g14411v3 BRADI_2g15950v3 BRADI_2g16660v3 BRADI_2g17200v3 BRADI_2g19540v3 BRADI_2g22140v3
BRADI_2g23040v3 BRADI_2g23250v3 BRADI_2g24230v3 BRADI_2g25613v3 BRADI_2g27900v3 BRADI_2g31576v3 BRADI_2g33360v3 BRADI_2g33682v3 BRADI_2g36000v3
BRADI_2g39890v3 BRADI_2g40942v3 BRADI_2g41117v3 BRADI_2g42227v3 BRADI_2g45460v3 BRADI_2g45840v3 BRADI_2g47530v3 BRADI_2g49280v3 BRADI_2g49660v3
BRADI_2g50300v3 BRADI_2g51240v3 BRADI_2g52470v3 BRADI_2g53020v3 BRADI_2g54570v3 BRADI_2g55490v3 BRADI_2g55960v3 BRADI_2g56190v3 BRADI_2g56490v3
BRADI_2g56500v3 BRADI_2g56710v3 BRADI_2g57220v3 BRADI_2g57230v3 BRADI_2g58835v3 BRADI_2g61767v3 BRADI_3g02640v3 BRADI_3g03532v3 BRADI_3g05530v3
BRADI_3g06107v3 BRADI_3g15890v3 BRADI_3g19893v3 BRADI_3g22980v3 BRADI_3g26930v3 BRADI_3g26967v3 BRADI_3g27800v3 BRADI_3g28450v3 BRADI_3g28700v3
BRADI_3g29700v3 BRADI_3g29797v3 BRADI_3g30380v3 BRADI_3g33544v3 BRADI_3g33850v3 BRADI_3g35000v3 BRADI_3g36370v3 BRADI_3g37047v3 BRADI_3g37260v3
BRADI_3g37790v3 BRADI_3g37820v3 BRADI_3g38220v3 BRADI_3g38670v3 BRADI_3g39440v3 BRADI_3g41490v3 BRADI_3g43890v3 BRADI_3g44947v3 BRADI_3g44950v3
BRADI_3g45030v3 BRADI_3g45206v3 BRADI_3g45940v3 BRADI_3g48240v3 BRADI_3g48590v3 BRADI_3g49400v3 BRADI_3g50110v3 BRADI_3g50420v3 BRADI_3g51600v3
BRADI_3g54200v3 BRADI_3g54267v3 BRADI_3g56610v3 BRADI_3g57525v3 BRADI_3g58980v3 BRADI_4g01710v3 BRADI_4g02925v3 BRADI_4g05450v3 BRADI_4g06370v3
BRADI_4g07390v3 BRADI_4g07911v3 BRADI_4g08420v3 BRADI_4g08550v3 BRADI_4g09120v3 BRADI_4g10998v3 BRADI_4g17336v3 BRADI_4g20537v3 BRADI_4g23800v3
BRADI_4g24320v3 BRADI_4g25430v3 BRADI_4g27376v3 BRADI_4g29506v3 BRADI_4g29540v3 BRADI_4g31070v3 BRADI_4g32941v3 BRADI_4g33899v3 BRADI_4g34056v3
BRADI_4g34980v3 BRADI_4g35170v3 BRADI_4g35301v3 BRADI_4g36690v3 BRADI_4g37460v3 BRADI_4g37770v3 BRADI_4g40310v3 BRADI_4g42710v3 BRADI_4g42890v3
BRADI_4g43220v3 BRADI_4g44530v3 BRADI_4g44760v3 BRADI_4g45410v3 BRADI_5g00590v3 BRADI_5g02037v3 BRADI_5g06390v3 BRADI_5g06628v3 BRADI_5g10090v3
BRADI_5g10140v3 BRADI_5g12452v3 BRADI_5g13610v3 BRADI_5g13930v3 BRADI_5g14550v3 BRADI_5g15300v3 BRADI_5g15396v3 BRADI_5g15835v3 BRADI_5g15970v3
BRADI_5g16100v3 BRADI_5g18290v3 BRADI_5g20410v3 BRADI_5g20890v3 BRADI_5g21757v3 BRADI_5g23150v3 BRADI_5g23158v3 BRADI_5g24850v3 BRADI_5g25450v3
BRADI_5g25547v3 BRADI_5g26907v3

Co-expression Network with WGCNA

1

1.

Enter ids and or select file for batch upload. Else enter ids or select file or list from workspace for comparing to a reference list.

Enter IDs:
[Supported IDs](#)

BRADI_1g01507v3 BRADI_1g03590v3 BRADI_1g03610v3 BRADI_1g04440v3 BRADI_1g07237v3 BRADI_1g08410v3 BRADI_1g08870v3 BRADI_1g09950v3 BRADI_1g11630v3 BRADI_1g11860v3 BRADI_1g13232v3 BRADI_1g14940v3 BRADI_1g15030v3 BRADI_1g16190v3 BRADI_1g16620v3 BRADI_1g20360v3 BRADI_1g20620v3 BRADI_1g20930v3 BRADI_1g26280v3 BRADI_1g27490v3 BRADI_1g29040v3 BRADI_1g30247v3 BRADI_1g32330v3 BRADI_1g32660v3 BRADI_1g32990v3 BRADI_1g33300v3 BRADI_1g35742v3 BRADI_1g36290v3 BRADI_1g37080v3 BRADI_1g37175v3 BRADI_1g37750v3 BRADI_1g38760v3 BRADI_1g42730v3 BRADI_1g44047v3 BRADI_1g44230v3 BRADI_1g45080v3 BRADI_1g45160v3 BRADI_1g45200v3 BRADI_1g45530v3 BRADI_1g47767v3 BRADI_1g48010v3 BRADI_1g50370v3 BRADI_1g51610v3 BRADI_1g53320v3 BRADI_1g53850v3 BRADI_1g55500v3 BRADI_1g55620v3 BRADI_1g61040v3 BRADI_1g61170v3 BRADI_1g64850v3 BRADI_1g65480v3 BRADI_1g66800v3 BRADI_1g67040v3 BRADI_1g67290v3 BRADI_1g68720v3 BRADI_1g69400v3 BRADI_1g70700v3 BRADI_1g72740v3 BRADI_1g72950v3 BRADI_1g73120v3 BRADI_1g74300v3 BRADI_1g76950v3 BRADI_1g77637v3 BRADI_2g00831v3 BRADI_2g02110v3 BRADI_2g02400v3 BRADI_2g05640v3 BRADI_2g07040v3 BRADI_2g12830v3 BRADI_2g13110v3 BRADI_2g13870v3 BRADI_2g13870v30 BRADI_2g13870v33 BRADI_2g14411v3 BRADI_2g15950v3 BRADI_2g16660v3 BRADI_2g17200v3 BRADI_2g19540v3 BRADI_2g22140v3 BRADI_2g23040v3 BRADI_2g23250v3 BRADI_2g24230v3 BRADI_2g25613v3 BRADI_2g27900v3 BRADI_2g31576v3 BRADI_2g33360v3 BRADI_2g33682v3 BRADI_2g36000v3 BRADI_2g39890v3 BRADI_2g40942v3 BRADI_2g41117v3 BRADI_2g42227v3 BRADI_2g45460v3 BRADI_2g45840v3 BRADI_2g47530v3 BRADI_2g49280v3 BRADI_2g49680v3 BRADI_2g50300v3 BRADI_2g51240v3 BRADI_2g52470v3 BRADI_2g53020v3 BRADI_2g54570v3 BRADI_2g55490v3 BRADI_2g55960v3 BRADI_2g56190v3 BRADI_2g56490v3 BRADI_2g56500v3 BRADI_2g56710v3 BRADI_2g57220v3 BRADI_2g57230v3 BRADI_2g58835v3 BRADI_2g61767v3 BRADI_3g02640v3 BRADI_3g03532v3 BRADI_3g05530v3 BRADI_3g06107v3 BRADI_3g15890v3 BRADI_3g22980v3 BRADI_3g26930v3 BRADI_3g26967v3 BRADI_3g27800v3 BRADI_3g28450v3 BRADI_3g28700v3 BRADI_3g29700v3 BRADI_3g29797v3 BRADI_3g30380v3 BRADI_3g33544v3 BRADI_3g33850v3 BRADI_3g35000v3 BRADI_3g36370v3 BRADI_3g37047v3 BRADI_3g37260v3 BRADI_3g37790v3 BRADI_3g3820v3 BRADI_3g38220v3 BRADI_3g38870v3 BRADI_3g39440v3 BRADI_3g41490v3 BRADI_3g43890v3 BRADI_3g44947v3 BRADI_3g44950v3 BRADI_3g45030v3 BRADI_3g45206v3 BRADI_3g45940v3 BRADI_3g48240v3 BRADI_3g48590v3 BRADI_3g49400v3 BRADI_3g50110v3 BRADI_3g50420v3 BRADI_3g51600v3 BRADI_3g54200v3 BRADI_3g54267v3 BRADI_3g56610v3 BRADI_3g57525v3 BRADI_3g58980v3 BRADI_4g01710v3 BRADI_4g02925v3 BRADI_4g05450v3 BRADI_4g06370v3 BRADI_4g07390v3 BRADI_4g07911v3 BRADI_4g08420v3 BRADI_4g08550v3 BRADI_4g09120v3 BRADI_4g10998v3 BRADI_4g17336v3 BRADI_4g20537v3 BRADI_4g23800v3 BRADI_4g24320v3 BRADI_4g25430v3 BRADI_4g27376v3 BRADI_4g29506v3 BRADI_4g29540v3 BRADI_4g31070v3 BRADI_4g32941v3 BRADI_4g33899v3 BRADI_4g34056v3 BRADI_4g34980v3 BRADI_4g35170v3 BRADI_4g35301v3 BRADI_4g36660v3 BRADI_4g37460v3 BRADI_4g37770v3 BRADI_4g40310v3 BRADI_4g42710v3 BRADI_4g42890v3 BRADI_4g43220v3 BRADI_4g44530v3 BRADI_4g44780v3 BRADI_4g45410v3 BRADI_5g00590v3 BRADI_5g02037v3 BRADI_5g06390v3 BRADI_5g06628v3 BRADI_5g10090v3 BRADI_5g10140v3 BRADI_5g12452v3 BRADI_5g13610v3 BRADI_5g13930v3 BRADI_5g14550v3 BRADI_5g15300v3 BRADI_5g15396v3 BRADI_5g15835v3 BRADI_5g15970v3 BRADI_5g16100v3 BRADI_5g18290v3 BRADI_5g20410v3 BRADI_5g20890v3 BRADI_5g21757v3 BRADI_5g23150v3 BRADI_5g23158v3 BRADI_5g24850v3 BRADI_5g25450v3 BRADI_5g25547v3 BRADI_5g26907v3

Upload IDs:
[File format](#)

Seleccionar archivo

 Ningún archivo seleccionado

Please [login](#) to be able to select lists from your workspace.

☒ ID List

☐ Previously exported text search results

☐ Workspace list

☐ PANTHER Generic Mapping

☐ ID's from Reference Proteome Genome

Organism for id list

 Abrus precatorius (ABRPR)

☐ VCF file

 Flanking region

20 Kb

☐ Search Enhancer Data

2.

Select organism.

Bacteroides thetaiotaomicron

Batrachochytrium dendrobatidis

Bos taurus

Brachypodium distachyon

Bradyrhizobium diazoefficiens

3.

Select Analysis.

☐ Functional classification viewed in gene list

☐ Functional classification viewed in graphic charts

☐ Bar chart

☐ Pie chart

☒ Statistical overrepresentation test

GO biological process complete

☐ Statistical enrichment test

submit

SELECT REFERENCE LIST ?

For a reference list, you may upload your own list (recommended) or choose from available whole genome lists.

Upload Reference List from flat file or Workspace

Select Organism: (Not applicable for Generic mapping file or Reference Proteome ids)

Bacillus subtilis

Bacteroides thetaiotaomicron

Batrachochytrium dendrobatidis

Bos taurus

Brachypodium distachyon

Upload list:

Please select list type...

☒ Gene, Transcript, Protein and Alternate ID

☐ PANTHER Generic Mapping File

☐ ID's from Reference Proteome Genome

Organism for id list

Abrus precatorius (ABRPR)

☐ VCF file

 Flanking region

20 Kb

Upload list:

 Seleccionar archivo Ningún archivo seleccionado [supported IDs](#)

Upload list

If there are redundant IDs, only the first will be used in the analysis.

Please [login](#) to be able to select lists from your workspace.

Use a Reference List that includes all genes from a whole genome

Default whole-genome lists:

Select list

Brachypodium distachyon genes

Bos taurus genes

Brachypodium distachyon genes

Bradyrhizobium diazoefficiens genes

Branchiostoma floridae genes

Brassica campestris genes

Brassica napus genes

Caenorhabditis briggsae genes

Candida albicans genes

Canis lupus familiaris genes

Selection Summary:

Analysis Type: PANTHER Overrepresentation Test (Released 20240807)

Annotation Version and Release Date: GO Ontology database DOI: 10.5281/zenodo.16423886 Released 2025-07-22

Analyzed List: Client Text Box Input (Brachypodium distachyon)

Change

Reference List: Brachypodium distachyon (all genes in database)

Change

Annotation Data Set:

GO biological process complete

 ?

Test Type: ☒ Fisher's Exact ☐ Binomial

Correction: ☒ Calculate False Discovery Rate ☐ Use the Bonferroni correction for multiple testing ? ☐ No correction

Launch analysis



EXTRA EXCERCISES

EXTRA EXCERCISES

Task 07. Normalize the raw gene expression counts obtained from single-end read mappings to a reference genome, converting them into RPKM and TPM values. There are four genes (A, B, C and D) and three samples (1, 2 and 3).

Gene	Sample 1			Sample 2			Sample 3		
	Raw	RPKM	TPM	Raw	RPKM	TPM	Raw	RPKM	TPM
A (2,000 bp)	15	$0.15 \cdot 10^6$	$0.32 \cdot 10^6$	18			35		
B (4,000 bp)	25			30			70		
C (1,000 bp)	10			12			20		
D (10,000 bp)	0			1			1		
Total reads	50			61			126		

$$RPKM_i \text{ or } FPKM_i = \frac{q_i}{\frac{l_i}{10^3} * \frac{\sum_j q_j}{10^6}} = \frac{q_i}{l_i * \sum_j q_j} * 10^9$$

e.g. GeneA.Sample1 = $\frac{15}{\frac{2000}{10^3} * \frac{50}{10^6}} = \frac{15}{2000 * 50} * 10^9 = 0.15 * 10^6 \text{ RPKM}$

$$TPM_i = \frac{q_i / l_i}{\sum_j (q_j / l_j)} * 10^6$$

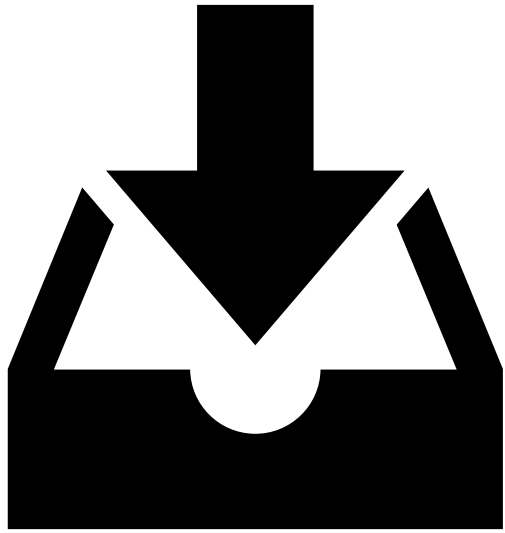
e.g. GeneA.Sample1 = $\frac{\frac{15}{2000}}{(\frac{15}{2000} + \frac{25}{4000} + \frac{10}{1000} + \frac{0}{10000})} * 10^6 = 0.32 * 10^6 \text{ TPM}$

EXTRA EXERCISES

Task 08. Using the compressed Fastqc file ABR3_D_03.fastq.gz (folder 01_RNAseq_raw), perform the steps of generating the QC report (FastQC software), filtering (Trimmomatic software) the sequences to remove adapters and low quality (and QC report after filtering) and aligning (Kallisto software) the filtered sequences against the reference transcriptome.

Task 09. Using Sleuth and the same input as in the previous example, visualize and compare the expression of the Bradi3g51200.1, Bradi1g37410.1, Bradi3g43870.1 transcripts using a heatmap. What differences do you observe when comparing their expression in the control group versus under drought conditions?

Task 10. Construct a co-expression network (WGCNA R package) of the samples and genes analyzed under the control condition (W). Compare the results of the modules obtained in both co-expression networks, drought (D) and control (W).



Deliverable Task

Deliverable Task

EXERCISE:

CONSTRUCT A GENE CO-EXPRESSION NETWORK (AND DEFINE THE COEXPRESSION MODULES) USING THE RNA-SEQ DATA FROM THE WATERED (W) SAMPLES AND ANSWER THE FOLLOWING QUESTIONS:

- (1)** How many samples and isoforms are included in the dataset "TPM_counts_Drought_W_dataset.csv"?
- (2)** How many samples are discarded after outlier analysis?
- (3)** What power value have you set as appropriate for calculating adjacency?
- (4)** How many co-expression modules are established before and how many after the module merging process?
- (5)** What is the hub isoform (or hub gene) of the cyan module?
- (6)** According to the module-trait association heat map, which module has the highest positive correlation with the "blwgrd (below ground biomass)" trait?

You must submit the solution (and files) to the exercise following these instructions:

- Create a folder called '**transcripts/**' in the same GitHub repository as session 1 and save all the relevant files created during the execution of the task there.
- Include a text file named "**Answers_transcript_YOURNAME.txt**" that displays the questions and the answers

TIPS:

- All the instructions, step by step, are included in the file "**Readme_deliverable_Task_transcript.txt**" available on Github.
- To answer the questions, run each chunk one by one and interpret the result obtained.

Thank you!

If you have any questions, please feel free to email me at
rsancho@eead.csic.es

